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Excel



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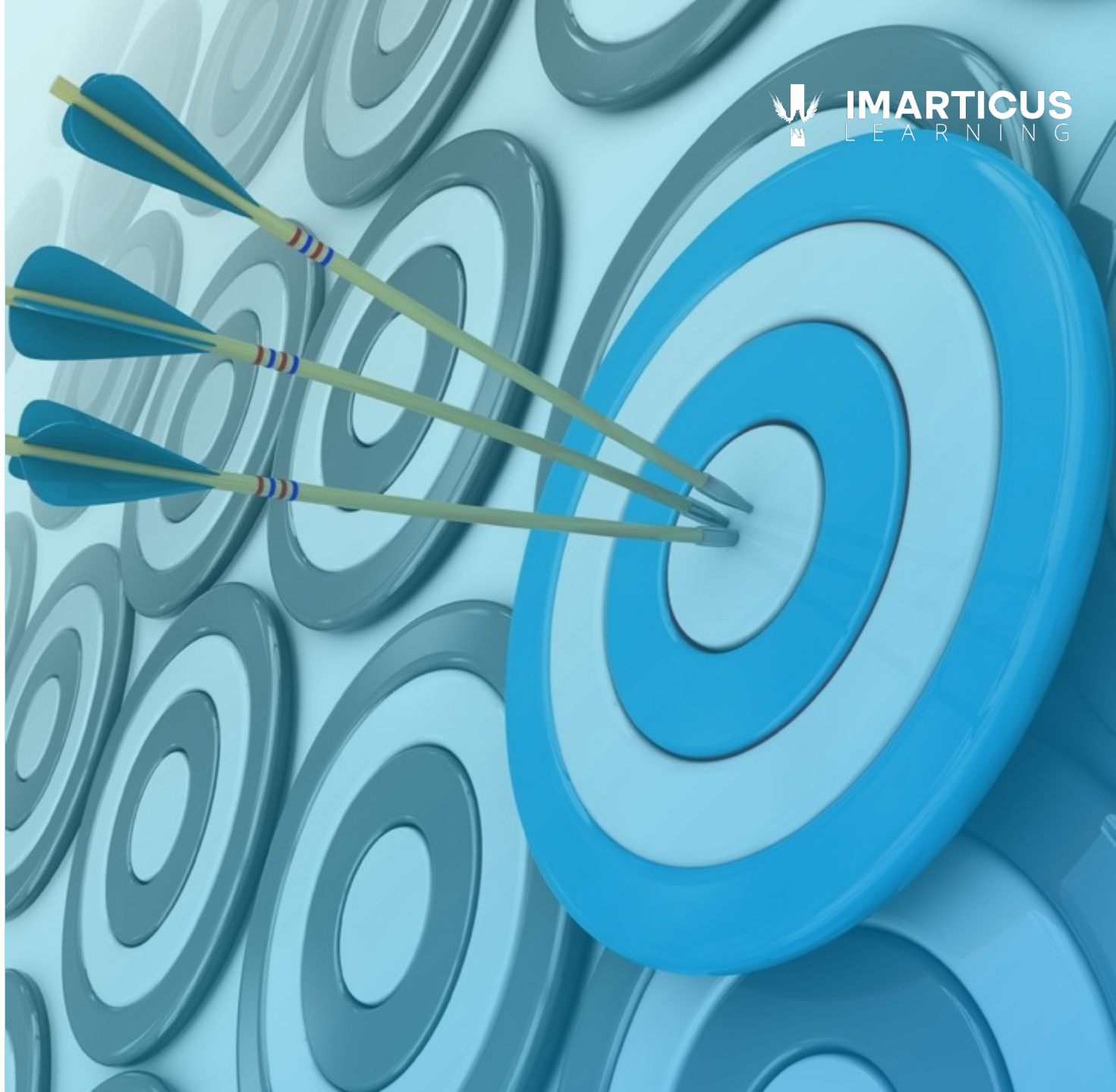
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LEARNING OBJECTIVES

At the end of this session, you will be able to:

- Perform different operations on the data
- Work upon various Excel functions
- Identify Various Excel Shortcuts
- Implement Sorting, Filtering and Other Data Operations
- Summarize Data with PivotTables and Tables Features
- Generate VLookup and HLookup
- Perform What-If Analysis
- Automate tasks using Macros



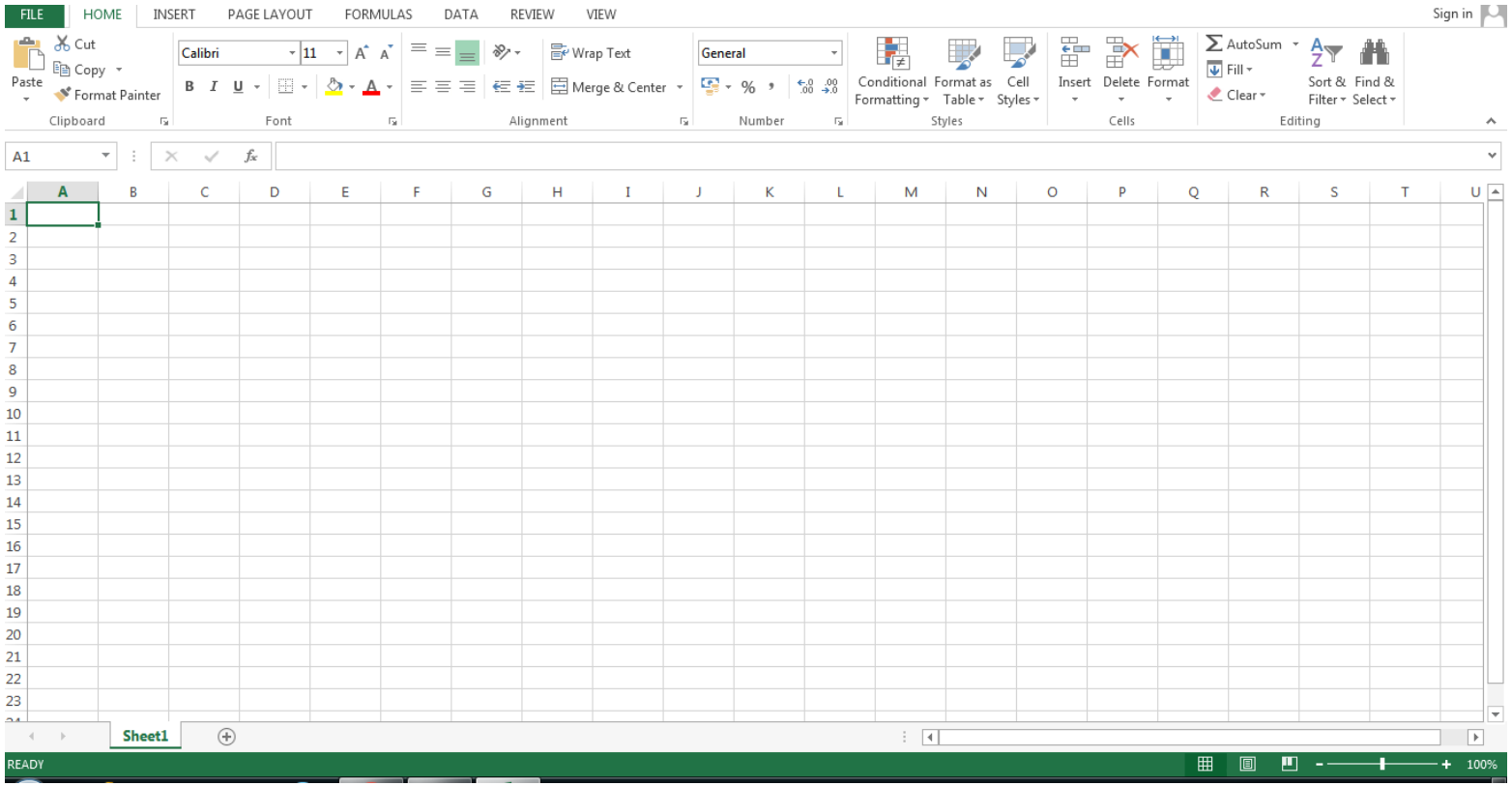
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Introduction to MS Excel



Introduction to MS Excel

- Microsoft Excel is a Microsoft Office product used to work upon electronic spreadsheets.
- It is a software application used to store, organize and analyze data.
- It was released in 1985 and ever since then it is a widely used tool to manipulate structured tabular data.



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Industry Uses of Excel



Applications of MS Excel

- Data Entry and Storage
- Collection and Verification of Business data
- Data Analysis and Interpretation
- Performing Calculations
- Accounting and Budgeting
- Report Generation
- Automating Repetitive Tasks
- Calendars and Scheduling
- Data Visualizations
- Data Forecasting





Advantages of Excel

Advantages of MS Excel

- **Easy To Store Data:** Since there is no limit to the amount of information that can be saved in a spreadsheet, MS Excel is widely used to save data or to analyse data. Filtering information in Excel is easy and convenient.
- **Easy To Recover Data:** If the information is written on a piece of paper, finding it may take longer, however, this is not the case with excel spreadsheets. Finding and recovering data is easy.
- **Application of Mathematical Formulas:** Doing calculations has become easier and less time consuming with the formulas option in MS excel
- **More Secure:** These spreadsheets can be password secured in a laptop or personal computer and the probability of losing them is way lesser compared to data written in registers or piece of paper.
- **Data at One Place:** Earlier, data was to be kept in different files and registers when the paperwork was done. Now, this has become convenient as more than one worksheet can be added in a single MS Excel file.
- **Neater and Clearer Visibility of Information:** When the data is saved in the form of a table, analysing it becomes easier. Thus, information is a spreadsheet that is more readable and understandable.



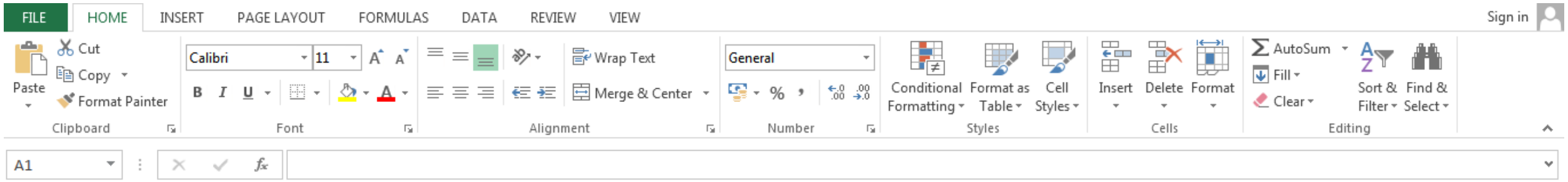
Features of Excel

Features of MS Excel

- **Home**
  Comprises options like font size, font styles, font colour, background colour, alignment, formatting options and styles, insertion and deletion of cells and editing options

- **Insert**
  Comprises options like table format and style, inserting images and figures, adding graphs, charts and sparklines, header and footer option, equation and symbols

- **Page Layout**
  Themes, orientation and page setup options are available under the page layout option



Features of MS Excel

- **Formulas**
  Since tables with a large amount of data can be created in MS excel, under this feature, you can add formulas to your table and get quicker solutions

- **Data**
  Adding external data (from the web), filtering options and data tools are available under this category

- **Review**
  Proofreading can be done for an excel sheet (like spell check) in the review category and a reader can add comments in this part

- **View**
  Different views in which we want the spreadsheet to be displayed can be edited here. Options to zoom in and out and pane arrangement are available under this category

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Terminologies in Excel



Terminologies in MS Excel

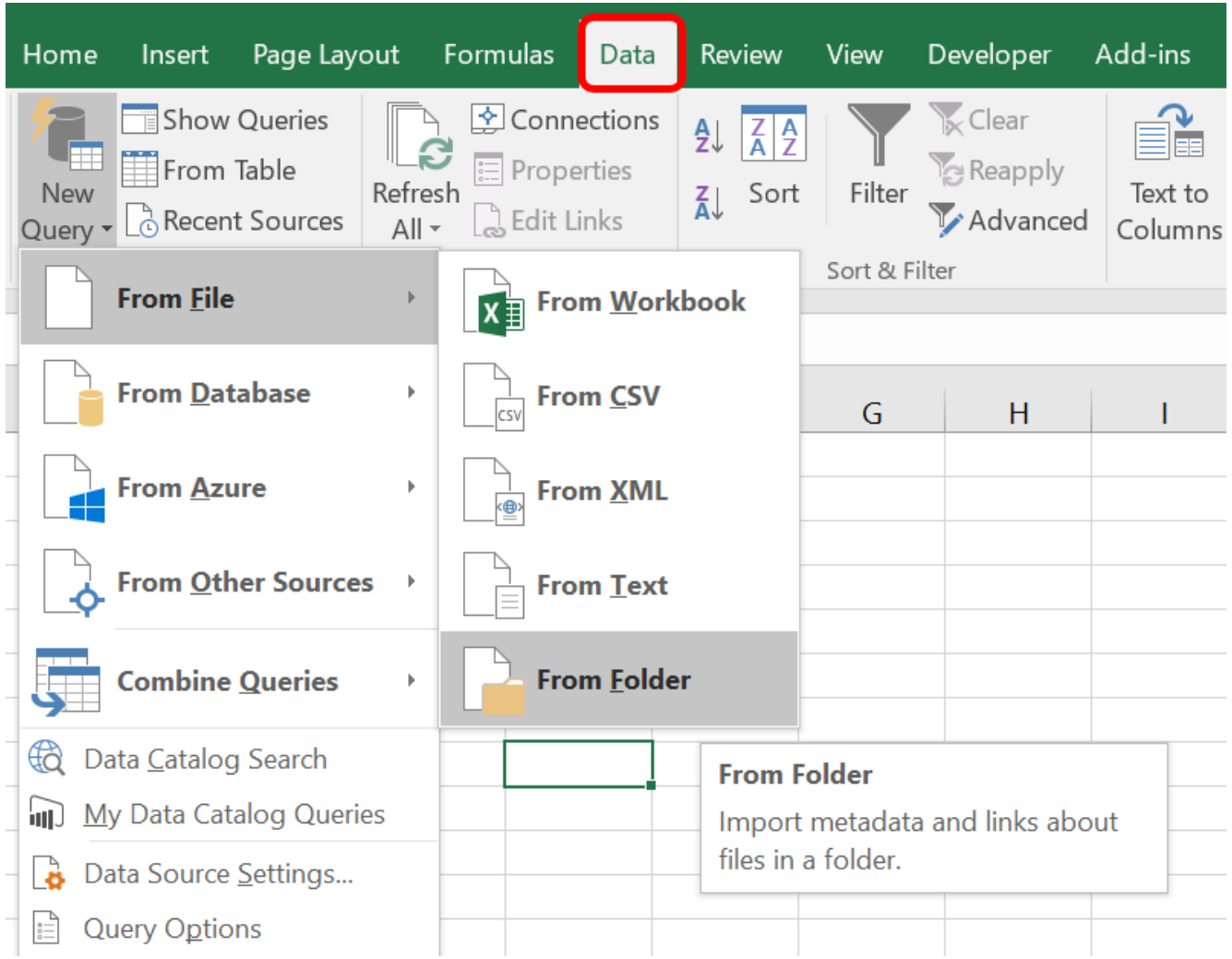
- **Cell-** A cell is a rectangle block/box present in a worksheet. Cells can be color-coded, show text, numbers, and the outcomes of calculations, based on the operations. An Active Cell is currently an open-cell for editing.
- **Workbook-** The workbook relates to an Excel spreadsheet record. It includes all the data that have been listed/sorted/calculated. A workbook that is available to be seen and edited by various users on a network is recognized as a Shared Workbook.
- **Worksheet-** Inside the workbook is where we get documents called worksheets. Also recognized as spreadsheets, a user can have multiple worksheets in a single workbook.
- **Workspace-** Very similar to worksheets in a workbook, a workspace enables users to open numerous files simultaneously.
- **Formula Bar-** The Formula Bar displays the data of an active cell. In the case of formulas, the formula bar will showcase all components of the formula.
- **Toolbar-** The toolbar that holds quick shortcuts like save, undo and redo, etc.
- **Chart-** An object that displays data visually / graphically.
- **Data Validation-** This feature of MS Excel helps to limit incorrect data from being inserted into the worksheet. Data validation aids consistency and precision in the data to be entered.
- **Ribbon-** Over the workbook is a section of command tabs named Ribbon. Several options are present behind every tab of the ribbon.



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Importing Data into Excel

Connecting to Different Data Sources

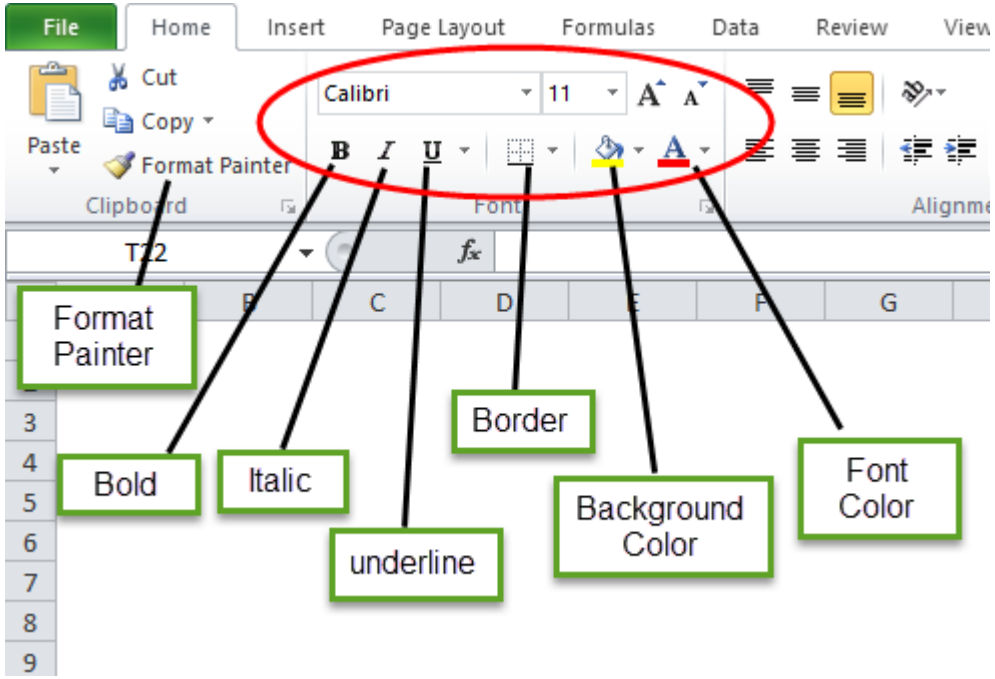


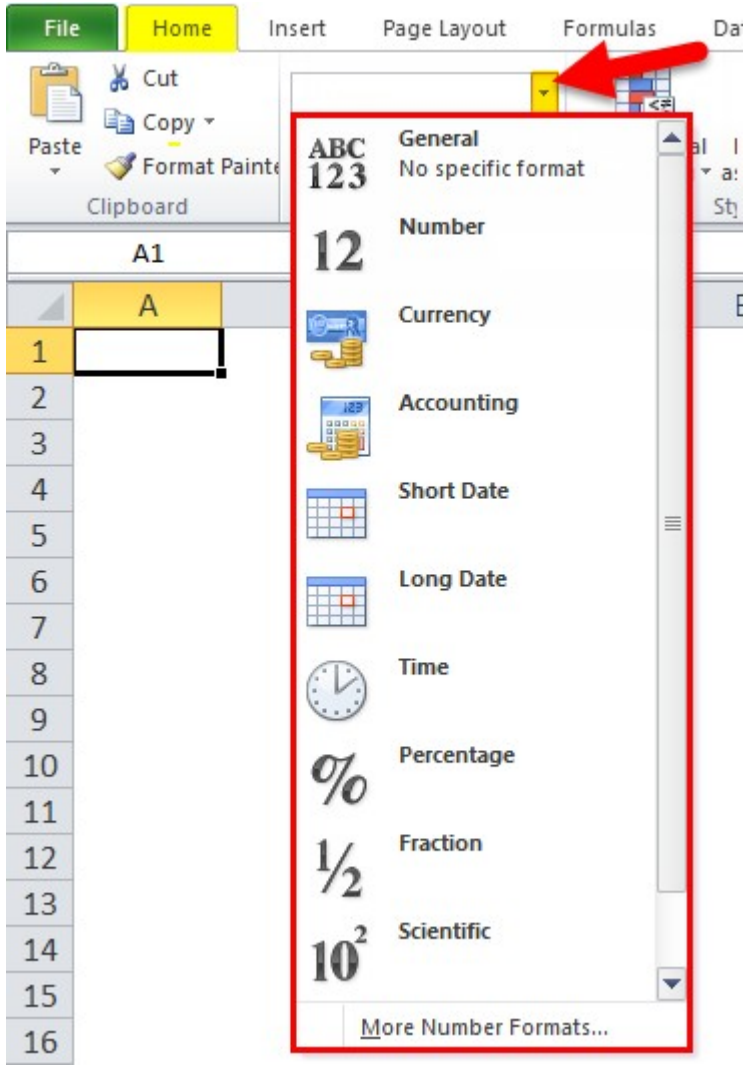
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Data Formatting in Excel

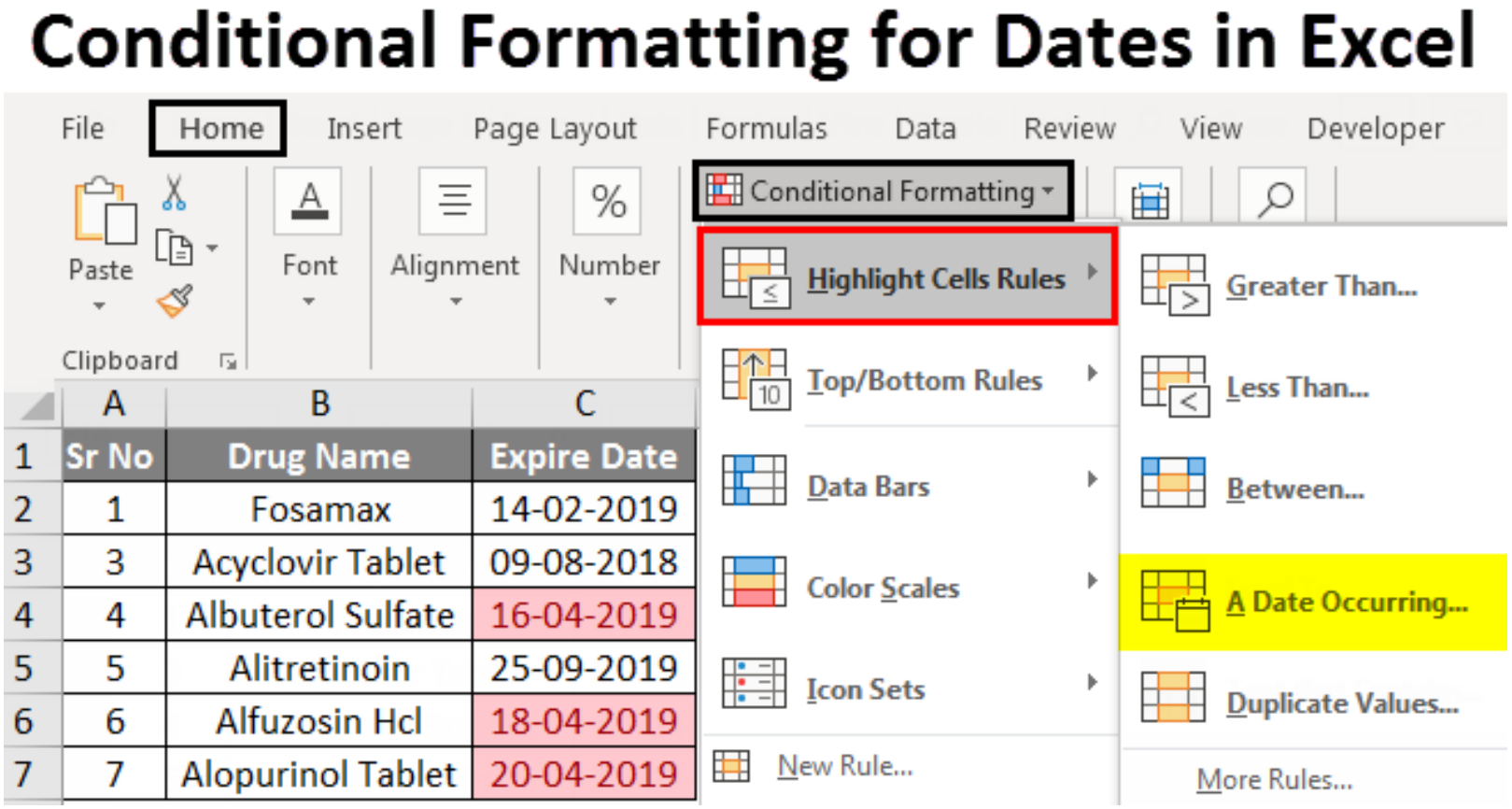


Different ways of Formatting Data





Conditional Formatting



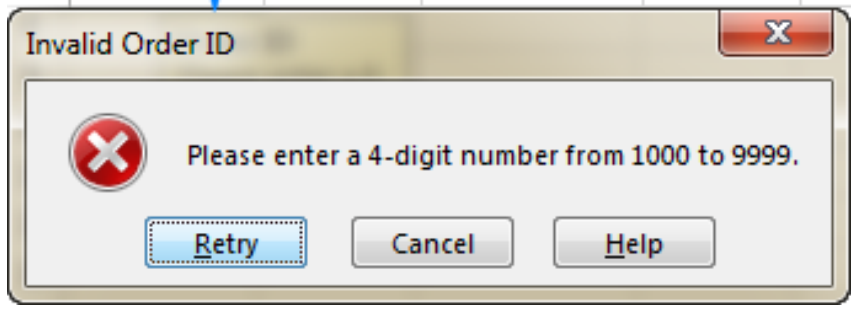
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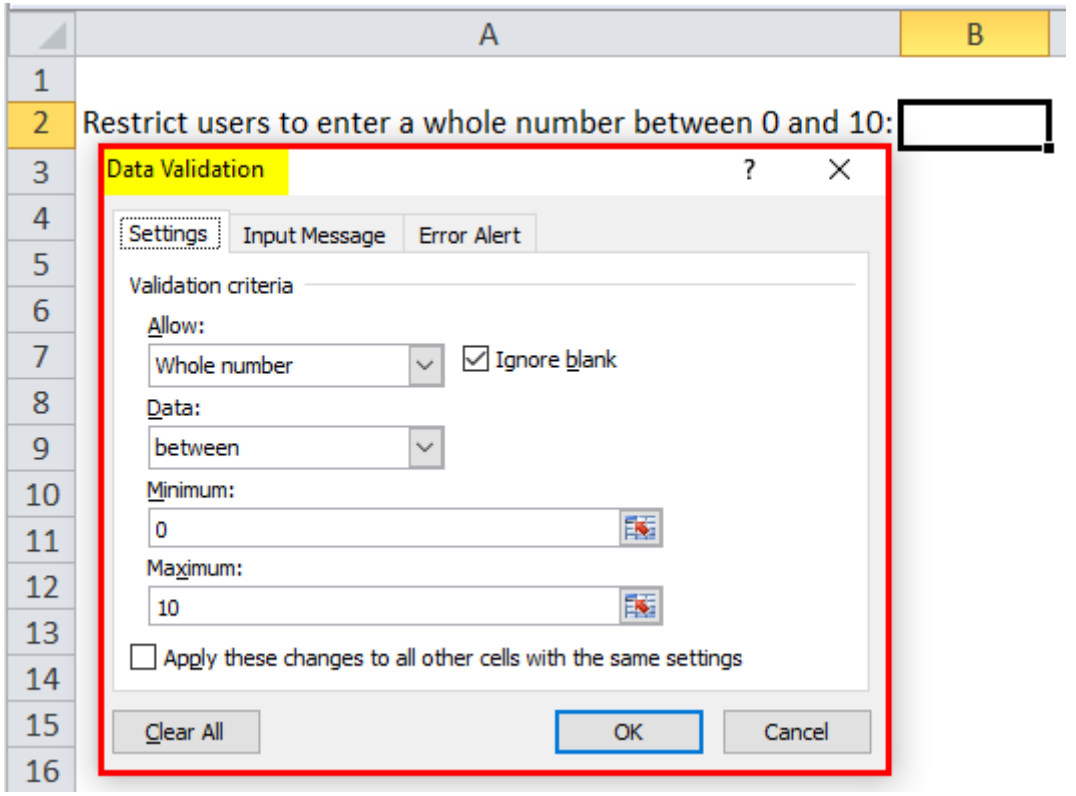
Data Validation in Excel



Data Validation

	A	B	C	D
1	Order ID	Item	Order date	Qty.
2	1001	Oranges	07/20/2017	30
3	1002	Bananas	07/20/2017	20
4	1003	Lemons	07/22/2017	10
5	1004	Cherries	07/22/2017	15
6	1005	Apples	07/24/2017	25
7	106			
8				
9				





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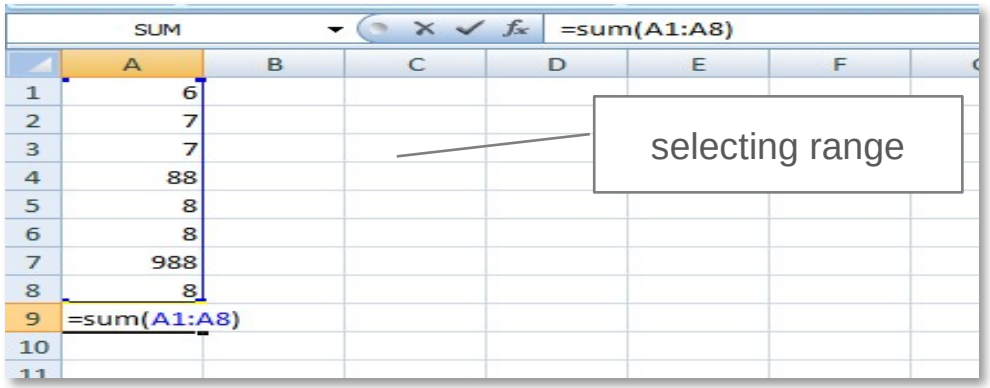
Performing Business Calculations using Various Excel Functions

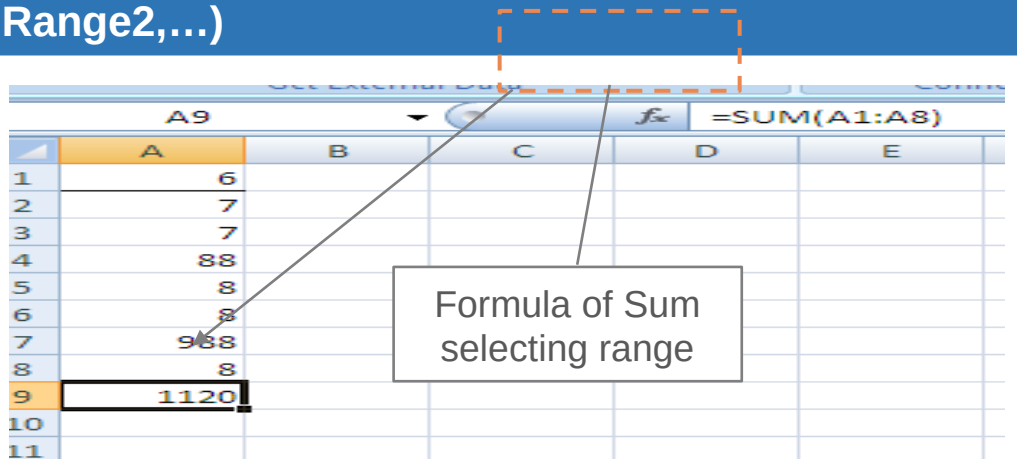


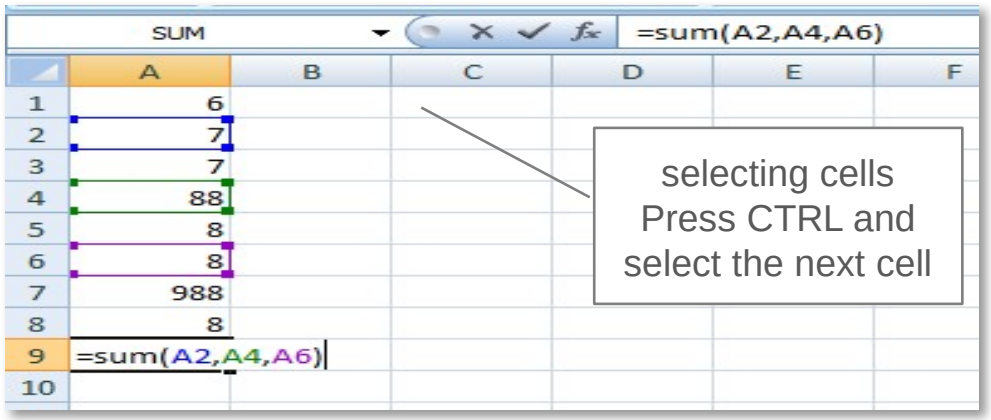
NUMERIC FUNCTIONS – SUM

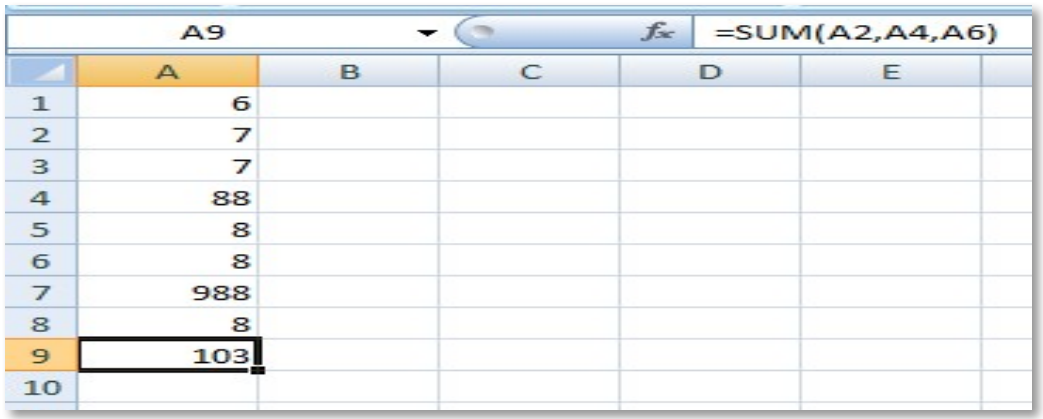
SUM Adds all the number in the range of cell. It also allows selection of individual cells instead of full range

Syntax -> =SUM(Range1,Range2,…)









NUMERICAL EXCEL FUNCTIONS - SUMIF & SUMIFS

Returns the Sum of cells specified by a given condition or Criteria(s)

Syntax

Syntax is **"=SUMIF(Range, Criteria, Sum Range)**

To Find sum of salaries of all Hunters:
=sumif(C2:C24,"Hunter",H2:H24)

Syntax is **"=SUMIFS(Sum Range, Criteria range 1, Criteria1, Criteria range 2, Criteria2,….)**

Note that Sum range here comes in beginning and for SumIF it comes in end

To Find sum of salaries of all Hunters with band E4:
=sumifs(H2:H24, C2:C24,"Hunter", D2:D24,"E4")

Further Modifications:

What if we need sum of salaries of all hunters with tenure more than 5 years?

Tip: Always use Sumifs, In this case you will not get confused with Format

NUMERICAL FUNCTIONS - COUNTIF & COUNTIFS

Returns the count of cells specified by a given condition or Criteria(s)

Syntax

Syntax is **“=COUNTIF(Range, Criteria)**

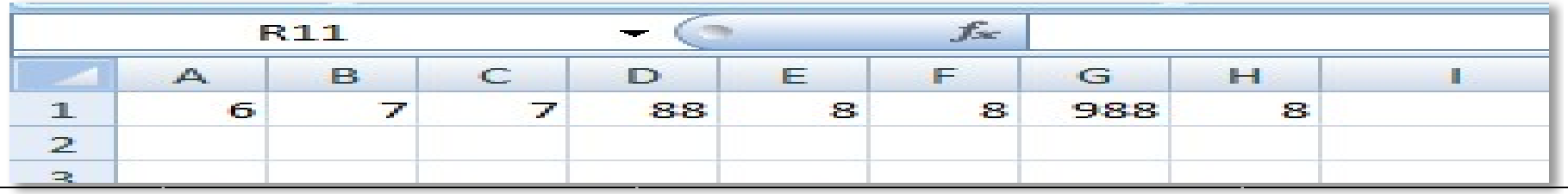
To Find count if of salaries of all Hunters:
=countif(C2:C24, “Hunter”,)

Syntax is **“=COUNTIFs(Criteria range 1, Criteria1, Criteria range 2, Criteria2,….)**

To Find count of salaries of all Hunters with band E4:
=countifs(C2:C24,”Hunter”, D2:D24,”E4”)

NUMERIC FUNCTIONS -MAX, MIN, AVERAGE, STDEV

Demonstration Example

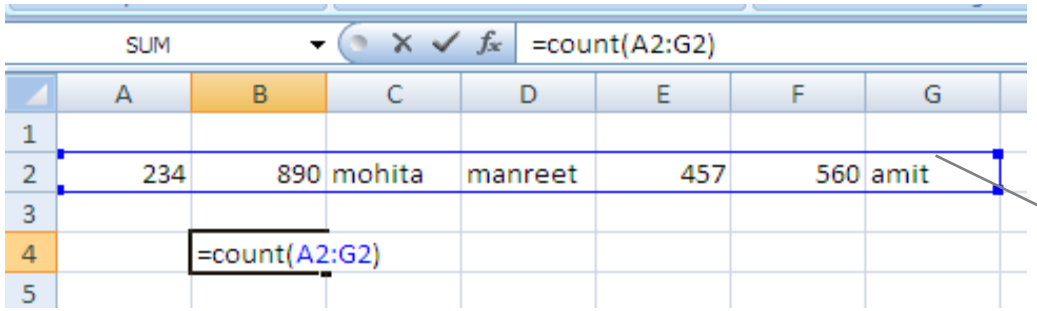


MAX() and MIN()	Returns the maximum or the minimum number from an array	=MAX or MIN(number1,number2,...)	=MAX(A1:H1) ***Output***- 988
AVERAGE()	Returns the simple average of the selected numbers	=AVERAGE(number1,number2,..)	=AVERAGE(A1:H1) ***Output***- 140
STDEV()	Returns the Standard deviation of the selected numbers	=STDEV(number1,number2,..)	=STDEV(A1:H1) ***Output***- 343.8052

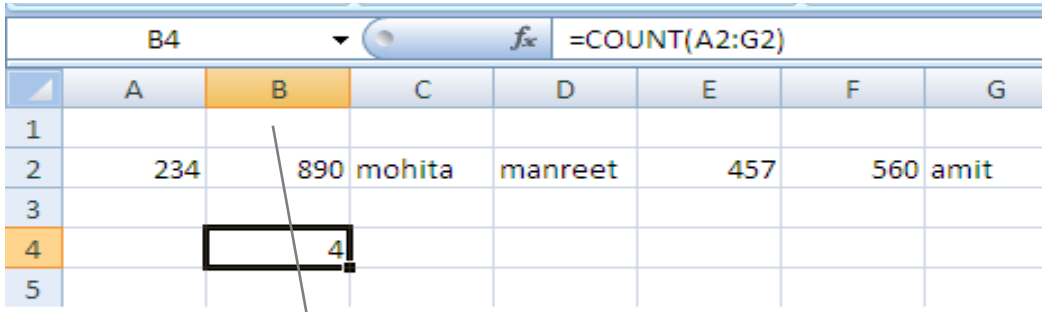
The above functions, function in similar manner as SUM wrt selection of specific cells and range

NUMERIC FUNCTIONS - COUNT

Function	Output	Syntax
COUNT()	Gives count of numbers in the range or discrete value mentioned in its parentheses	=COUNT(value1,value2, value3…)



Range with in which count of numbers is to be calculated



Calculated count

OTHER KEY CHARACTER FUNCTIONS

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Function	Usage
Rank	Provides the rank a value inside an array
Round	Returns the Rounded off value of a numerical variable
Int	Returns the integer part of a numerical variable
Ceil	Returns the ceiled value (next integer) of a numerical variable
Floor	Returns the floored value (pervious integer) of a numerical variable
Mod	Returns the remainder when a number is divided by a divisor
CountA	Counts the number of non blank values in an array
CountBlank	Counts the number of blank values in an array

LOGICAL FUNCTIONS – IF,AND, OR AND NOT

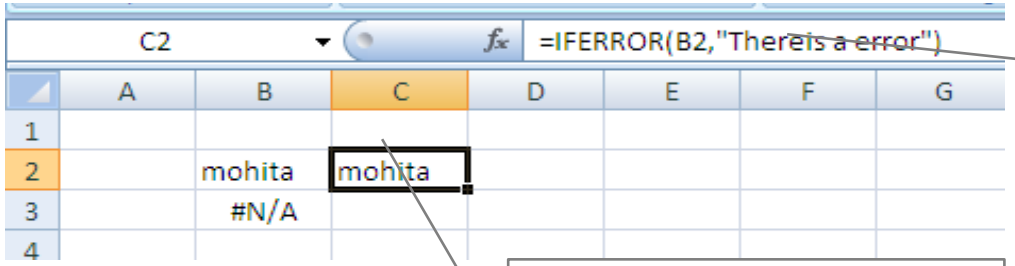
Demonstration Example

1	1	2		FALSE	TRUE		
2	2	4					
3	3	6					
4	4	8					
5	5	10					
6	6	12					
7	7	14					
8	8	16					
9	9	18					
10	10	20					

=AND(logical1,logical2,…)	Returns TRUE if all its arguments are TRUE; returns FALSE if one or more argument is FALSE	
=OR(logical1,logical2,…)	Returns TRUE if any argument is TRUE; returns FALSE if all arguments are FALSE	
=NOT(logical)	Reverses the value of its argument	
=IF(Condition,Value_if_True,Value_if_false)	Returns the Value_if_True if the condition is met Else it returns the Value_if_False	

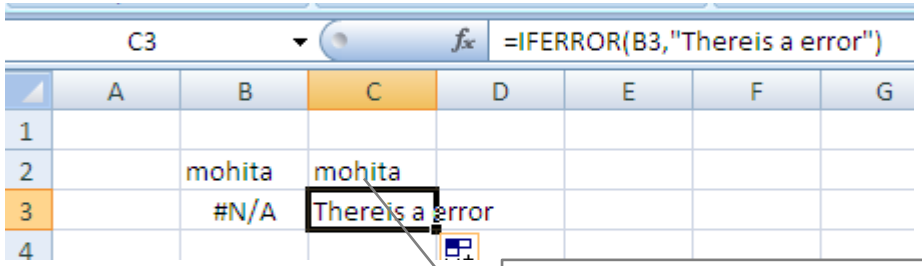
LOGICAL FUNCTIONS - IFERROR

Function	Output	Syntax
IFERROR()	IFERROR is equivalent to IF+ISERROR.	IFERROR(Value, Value_if_error)



Formula used

The function outputs “value” when no error is found



The function outputs “value_if_error” when error is found

DATE AND TIME FUNCTIONS - DATE

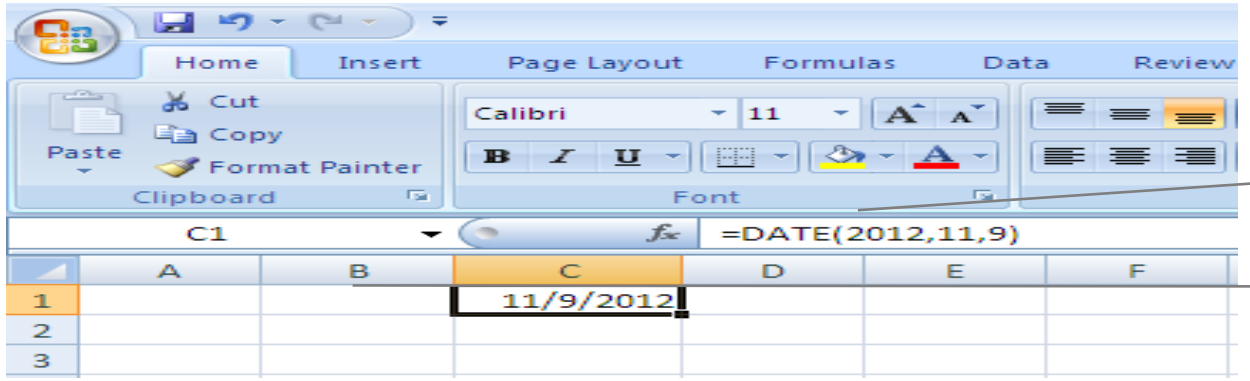
- It can be used to combine date elements from different locations.
- It can be used to convert dates to the computer's serial date.
- It can be used to ensure that dates used in calculations are entered correctly into a cell - to ensure they are treated as a date or a number instead of text.



DATE AND TIME FUNCTIONS - DATE

Syntax

- Syntax or formula for entering a date is “= DATE(Year, Month, Day) “
- Year - enter the year as a four digit number or the cell reference where it is located.
- Month - enter the year as a two digit number or the cell reference where it is located.
- Day - enter the day as a two digit number or the cell reference where it is located.



This is how date formula is used

This is how output date looks like

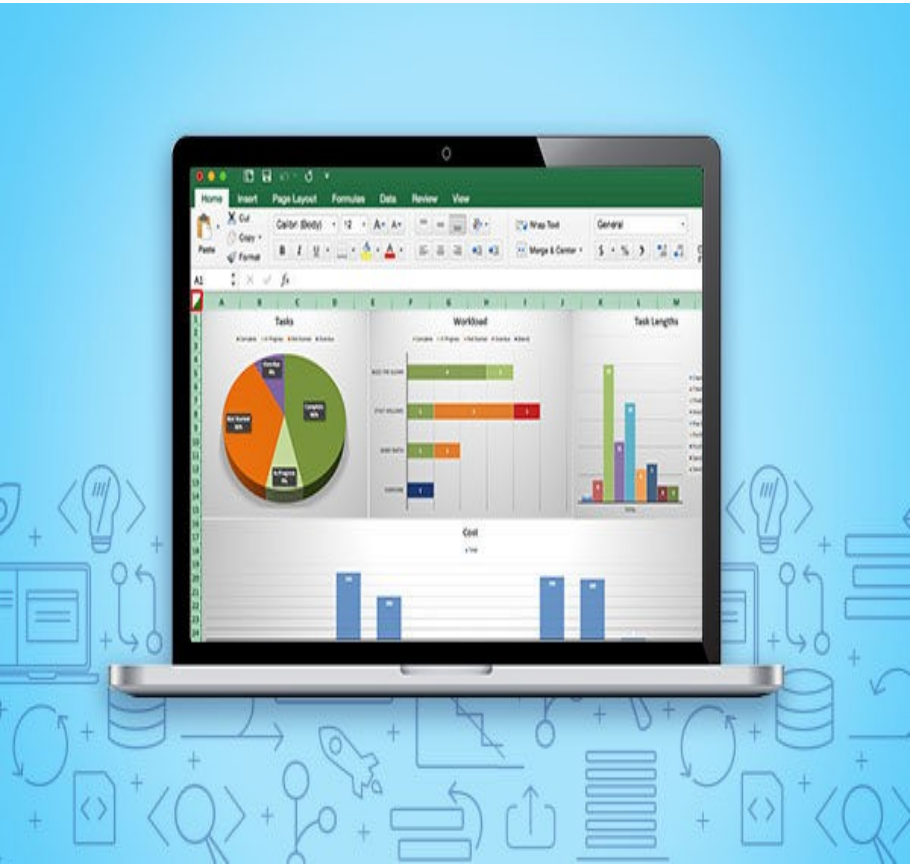
DATE AND TIME FUNCTIONS - TIME



- Returns the decimal number for a particular time.
- If the cell format was **General** before the function was entered, the result is formatted to Custom.
- The decimal number returned by TIME is a value ranging from 0 (zero) to 0.99999999, representing the times from 0:00:00 (12:00:00 AM) to 23:59:59 (11:59:59 P.M.).

DATE AND TIME FUNCTIONS - TIME

Syntax



Syntax is **“=TIME(hour,minute,second)”**

- **Hour** is a number from 0 (zero) to 32767 representing the hour. Any value greater than 23 will be divided by 24 and the remainder will be treated as the hour value.
  For example, TIME(27,0,0) = TIME(3,0,0) =.125 or 3:00 AM.
- **Minute** is a number from 0 to 32767 representing the minute. Any value greater than 59 will be converted to hours and minutes.
  For example, TIME(0,750,0) = TIME(12,30,0) =.520833 or 12:30 PM.
- **Second** is a number from 0 to 32767 representing the second. Any value greater than 59 will be converted to hours, minutes, and seconds.
  For example, TIME(0,0,2000) = TIME(0,33,20) =.023148 or 12:33:20 AM

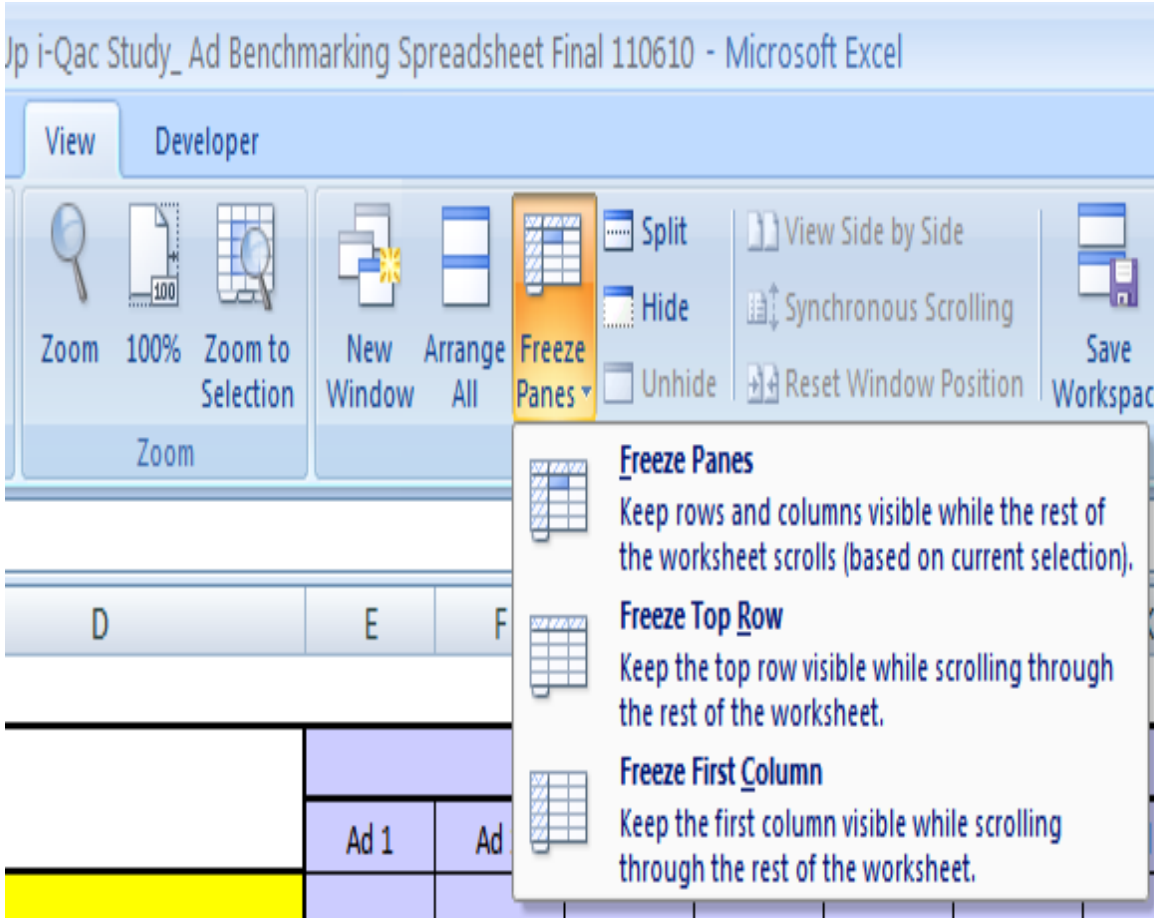


Sorting, Filtering and Other data operations

FREEZE PANES

Freeze Panes utility is used to lock rows or columns in place while you scroll through a worksheet.

- It is a very useful tool when you are looking across a large number of rows or columns, all of which won't fit into the screen.
- The header row/ column can be locked in place using Freeze panes tool to keep it visible while the rest of the worksheet can be scrolled through



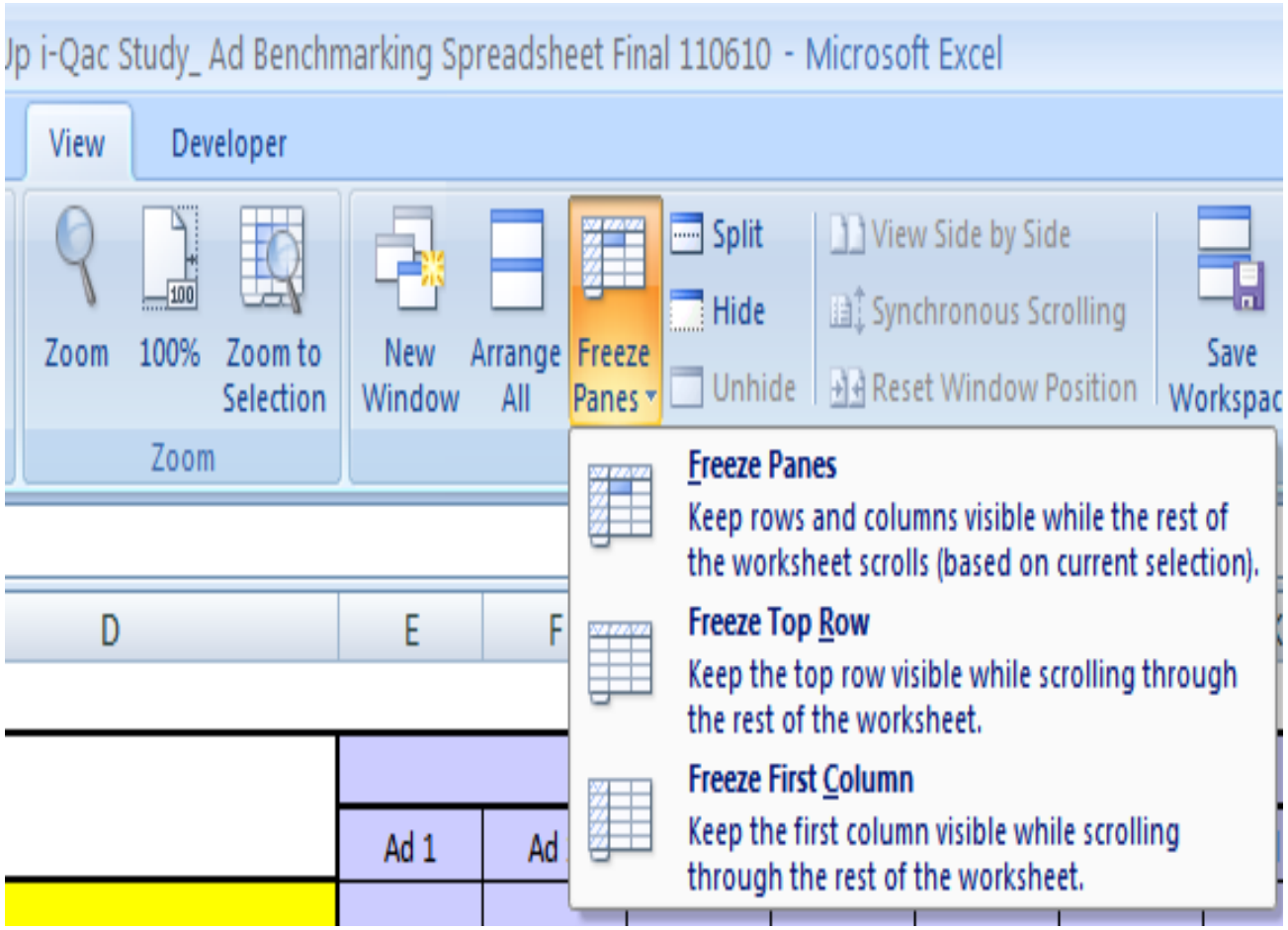
FREEZE PANES

Three variations are:

- Freeze Panes
- Freeze Top Row
- Freeze First Column

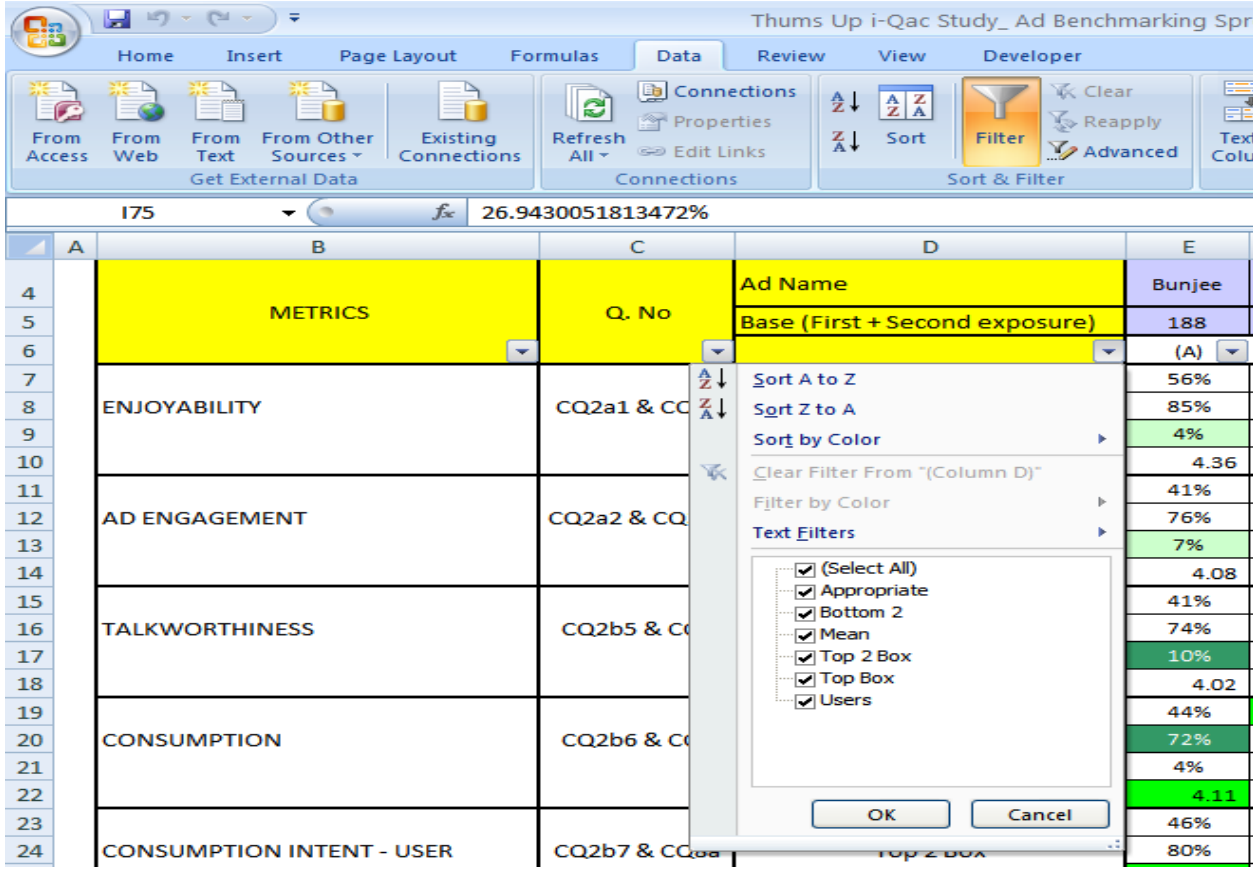
Tips:

- Try using ALT+W+F+F as a shortcut for the adjacent
- Use the same shortcut to get rid of the filters



FILTERS AUTO & ADVANCED

- Filtering is a quick and easy way to find and work with a subset of data in a range.
- A filtered range displays only the rows that meet the criteria you specify for a column. Microsoft Excel provides two commands for filtering ranges:
  - **AutoFilter,** which includes filter by selection, for simple criteria
  - **Advanced Filter,** for more complex criteria

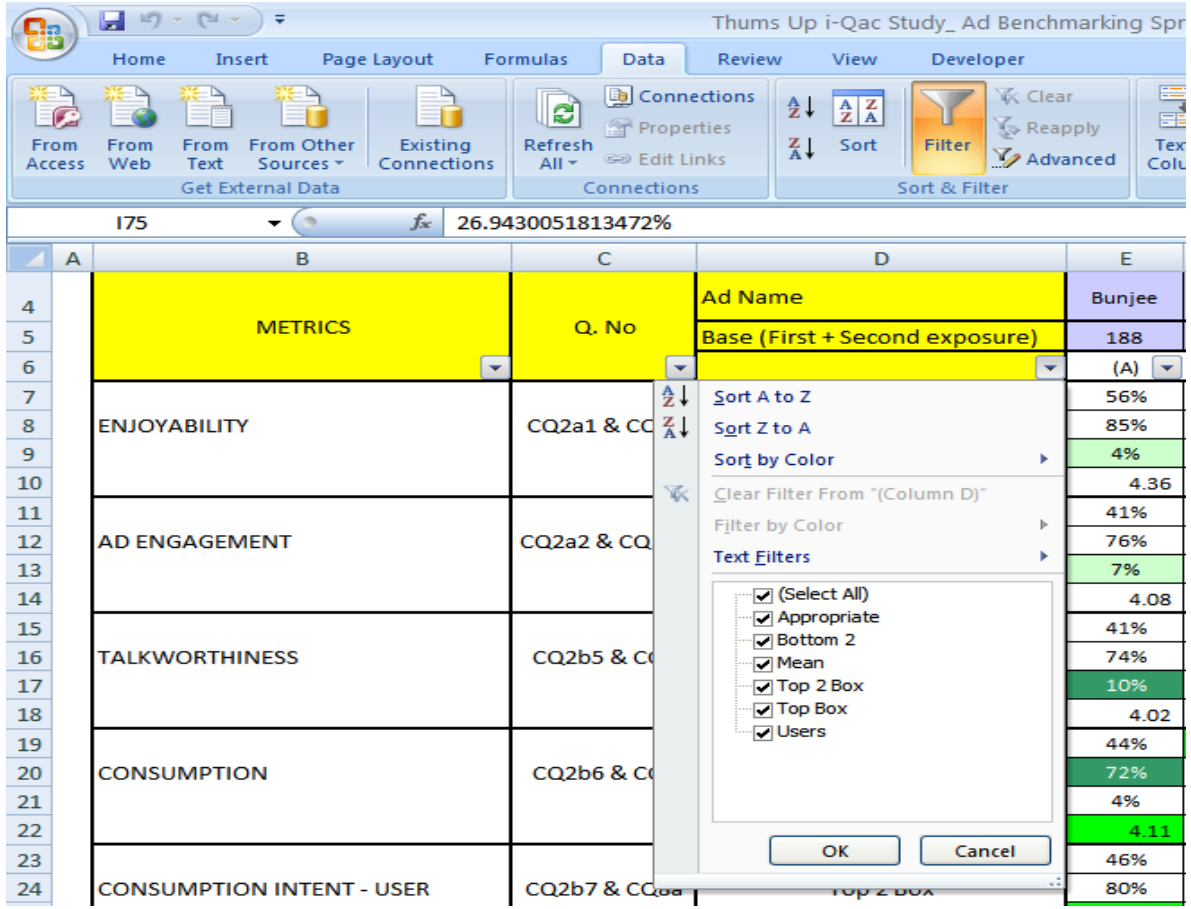


FILTERS AUTO & ADVANCED

- On the '**Data**' tab, click '**Filter**'
- Select '**AutoFilter**' or '**Advanced Filter**' as per requirement

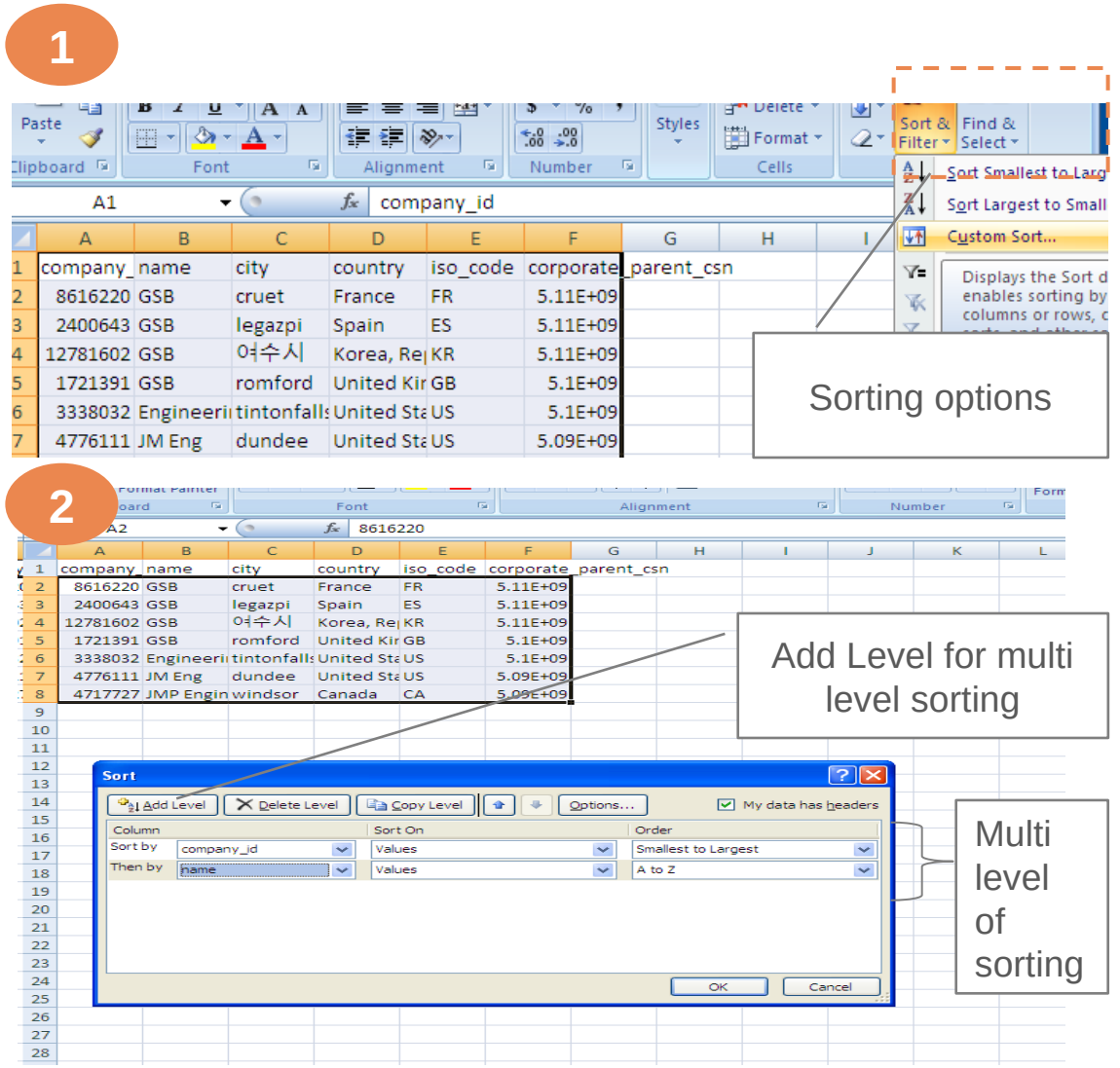
Tips:

- Try using ALT+A and then T/Q as a shortcut for the adjacent
- Use the same shortcut to get rid of the filters



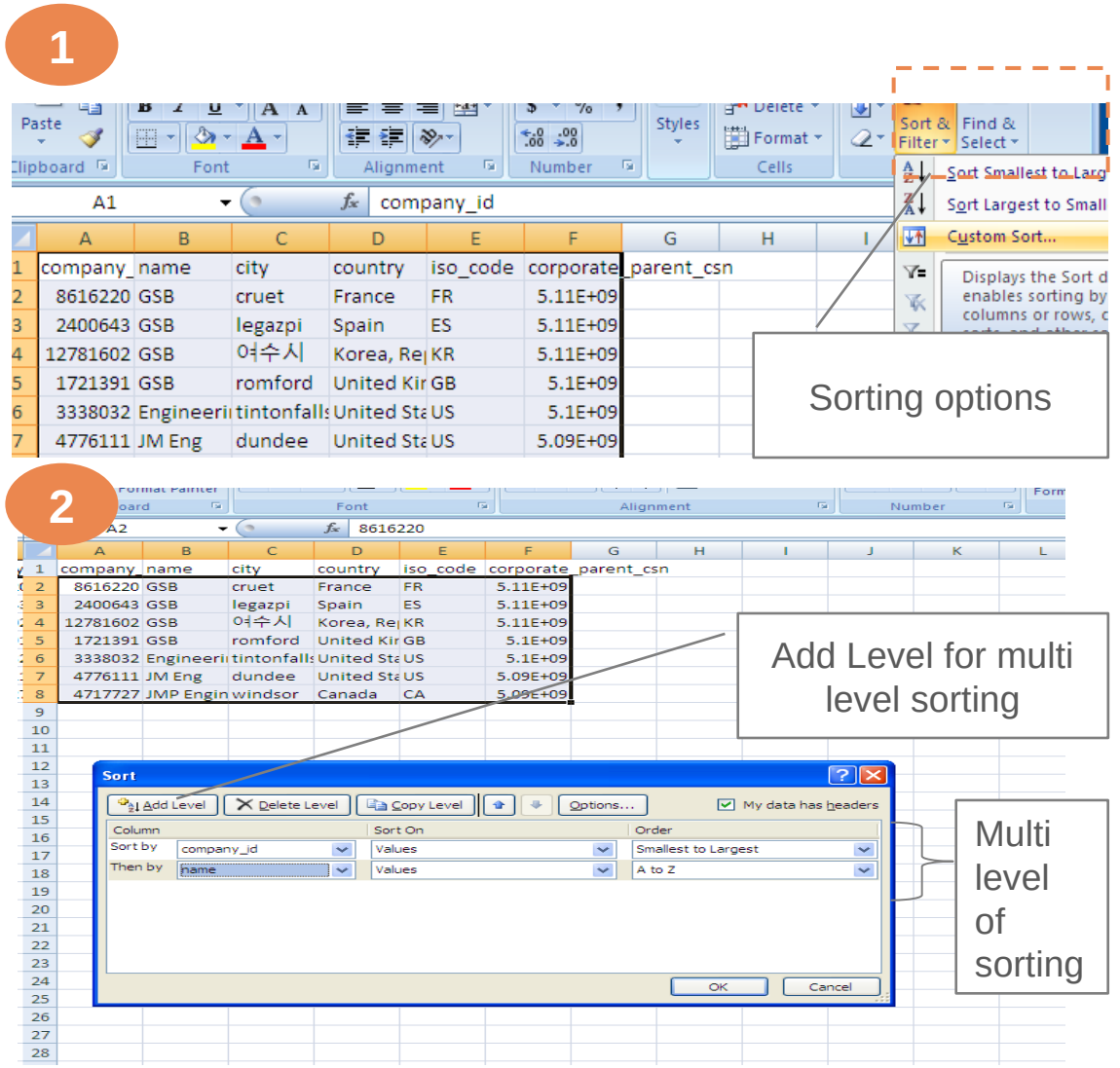
SIMPLE AND MULTI LEVEL SORTING

- Choose the column that you want to use to sort data in the excel sheet.
- Go to **Data** and choose '**Sort**' (ALT A + S), select either '**Sort Smallest to Largest**' or '**Sort Largest to Smallest**' or '**Custom sort**'



SIMPLE AND MULTI LEVEL SORTING(Cont…)

- In case the as a result of sorting a column, the records corresponding to it in other columns also have to be sorted, Select the entire table range, Select **Custom Sort** in Sort menu of Data tab and then specify the column by which table has to be sorted
- Multi level sorting leads to sorting on multiple criterion, keep on adding levels in custom sort window



TEXT TO COLUMNS

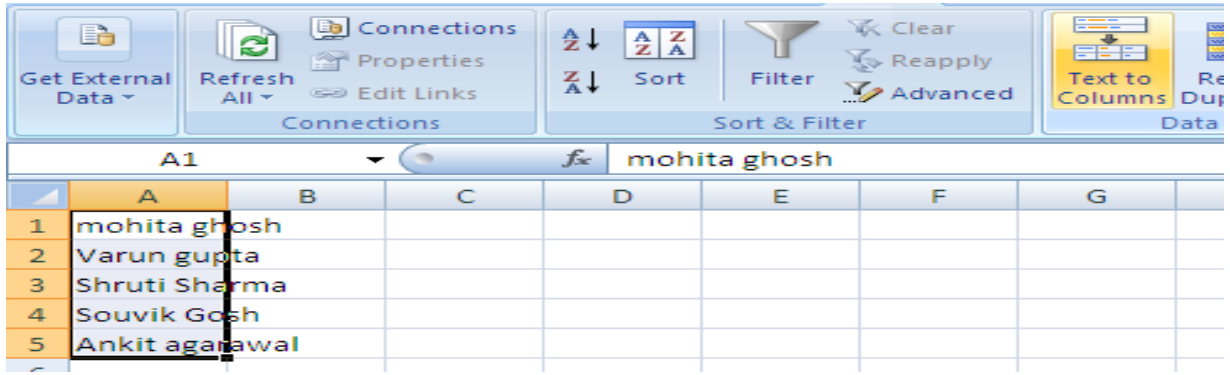
Text to Columns Wizard helps to separate simple cell content into multiple cells specified by the delimiter, such as first names and last names which are separated by space, into different columns as shown in the example (Next slide)

- Select the data array where Text to Columns is to be applied
- On the **Data** ribbon, click on Text to Columns tool (Alt+A+E)
- Select “Delimited” or “Fixed width” based on the type of data

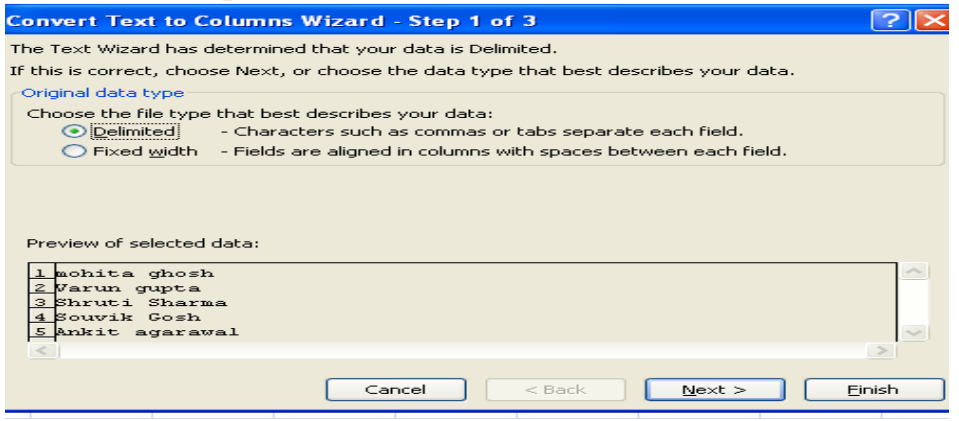


TEXT TO COLUMNS(Cont…..)

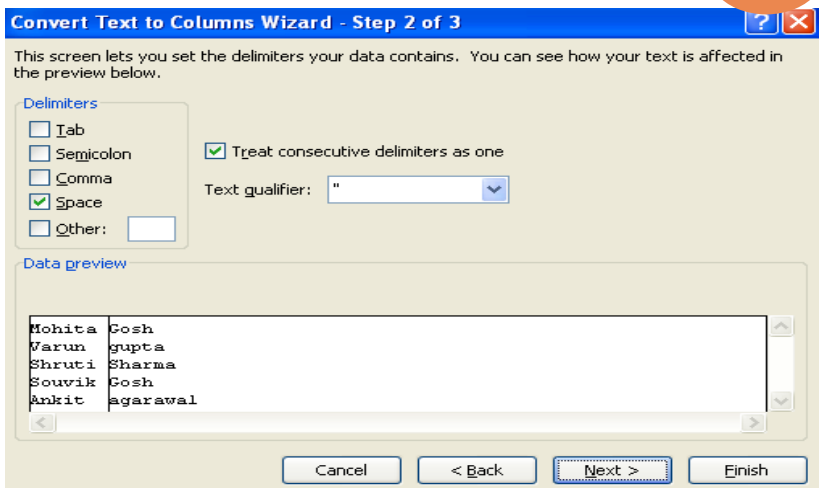
1



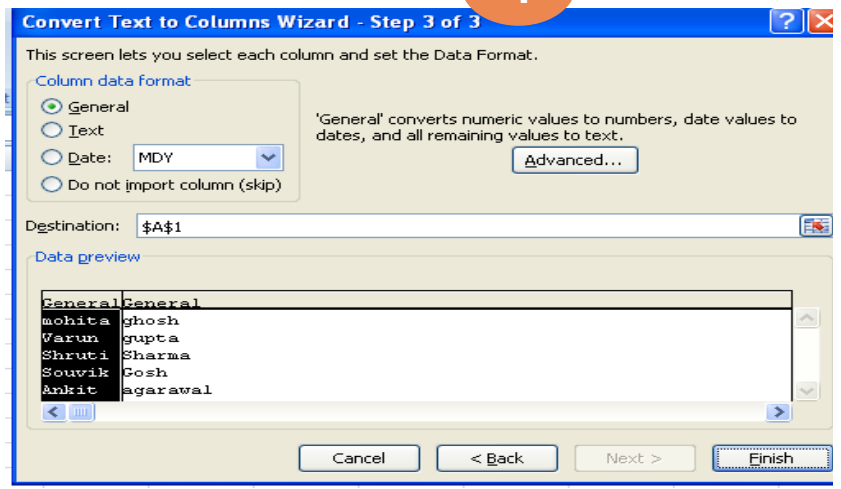
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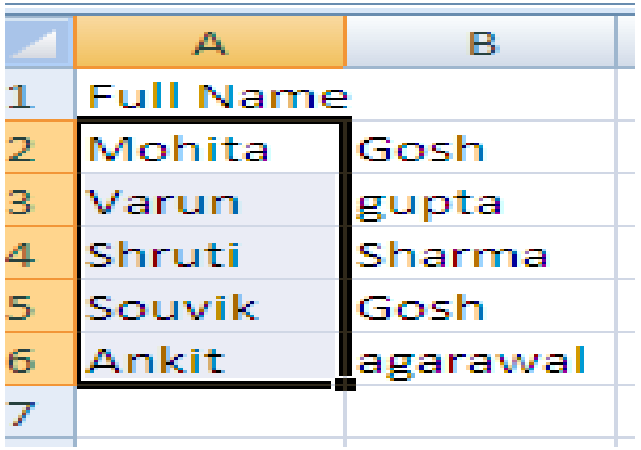
3



4



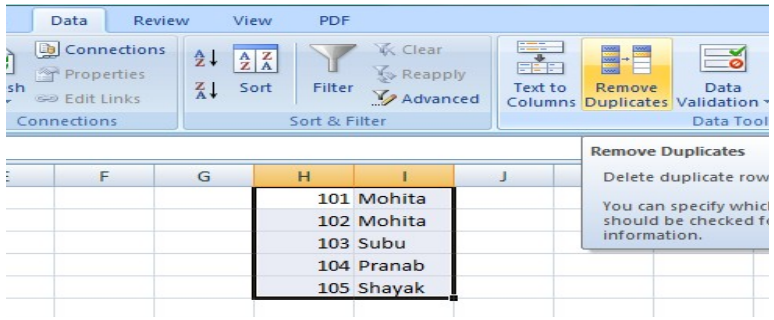
Output



REMOVE DUPLICATES

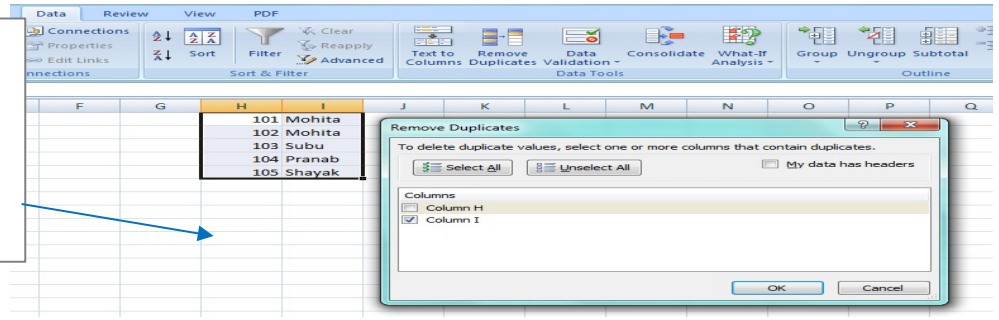
- This tool is used to remove duplicate or repeating values from data arrays arranged in columns. Effective in creating unique value lists
- Select the relevant columns, click on “Remove Duplicates” on the data ribbon and follow instructions (Alt+A+M)

1

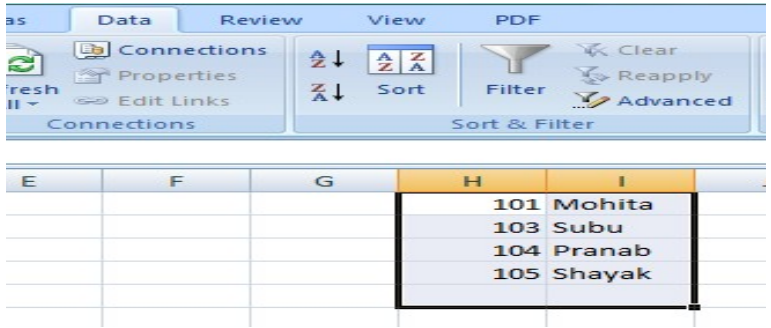


2

Select the column based on which duplicates have to be removed



3

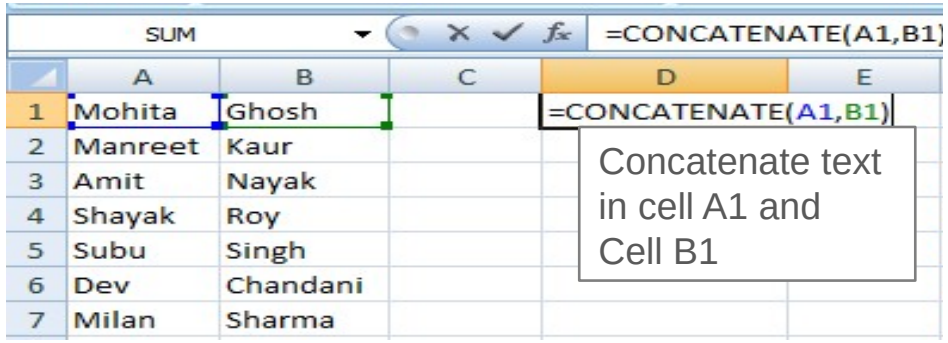


- In Column I ‘Mohita’ was repeated and hence removed
- If Column H and I both were selected then no rows would have been removed

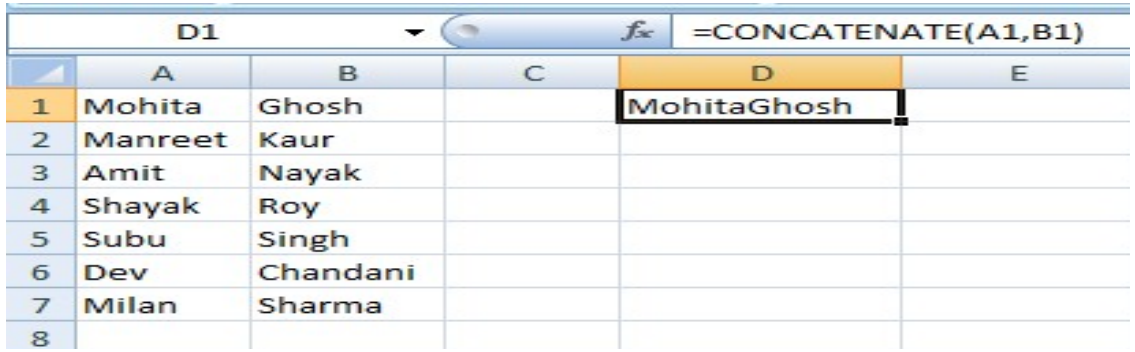
CHARACTER FUNCTIONS - CONCATENATE

Function	Output	Syntax
CONCATENATE()	Combines text from multiple columns in a single column	CONCATENATE (text1,text2,...)

Concatenation using CONCATENATE function

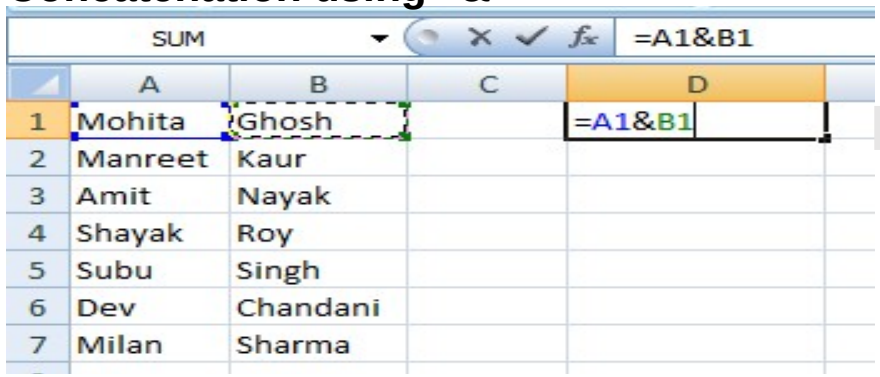


	A	B	C	D	E
1	Mohita	Ghosh			
2	Manreet	Kaur			
3	Amit	Nayak			
4	Shayak	Roy			
5	Subu	Singh			
6	Dev	Chandani			
7	Milan	Sharma			



	A	B	C	D	E
1	Mohita	Ghosh		MohitaGhosh	
2	Manreet	Kaur			
3	Amit	Nayak			
4	Shayak	Roy			
5	Subu	Singh			
6	Dev	Chandani			
7	Milan	Sharma			
8					

Concatenation using "&"



	A	B	C	D
1	Mohita	Ghosh		
2	Manreet	Kaur		
3	Amit	Nayak		
4	Shayak	Roy		
5	Subu	Singh		
6	Dev	Chandani		
7	Milan	Sharma		

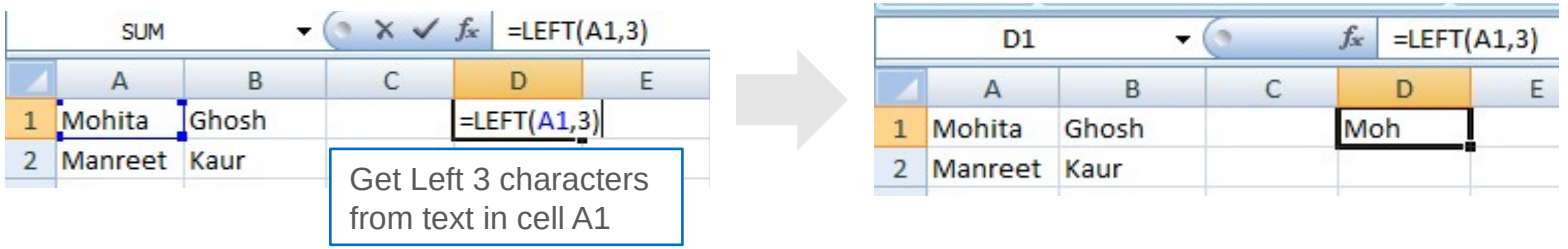
Tips:

1. Use CONCATENATE (text1," ",text2,) if you need space between two column entries. " " is used to insert any static text or space while concatenating
2. You can also use '&' instead of CONCATENATE

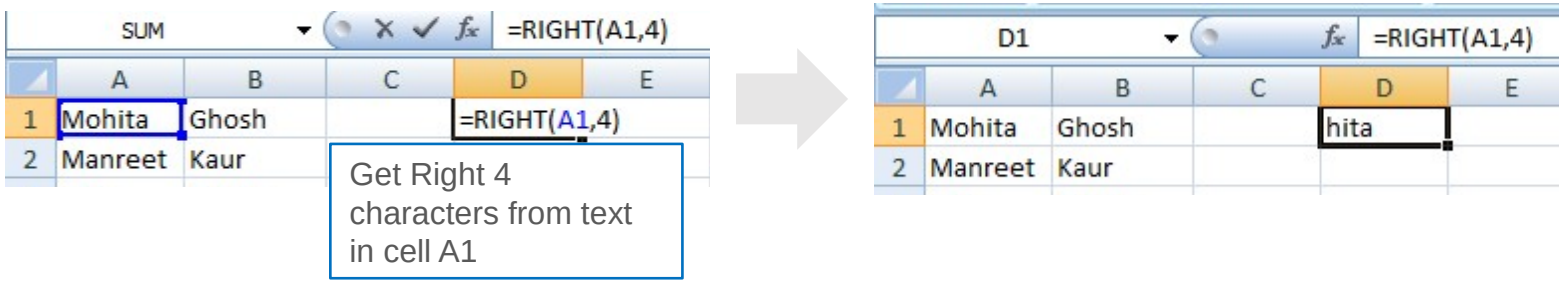
CHARACTER FUNCTIONS - LEFT, RIGHT

Function	Output	Syntax
LEFT() and RIGHT()	Returns the left or right characters in a text string	=LEFT or RIGHT(text,num_chars)

LEFT



RIGHT

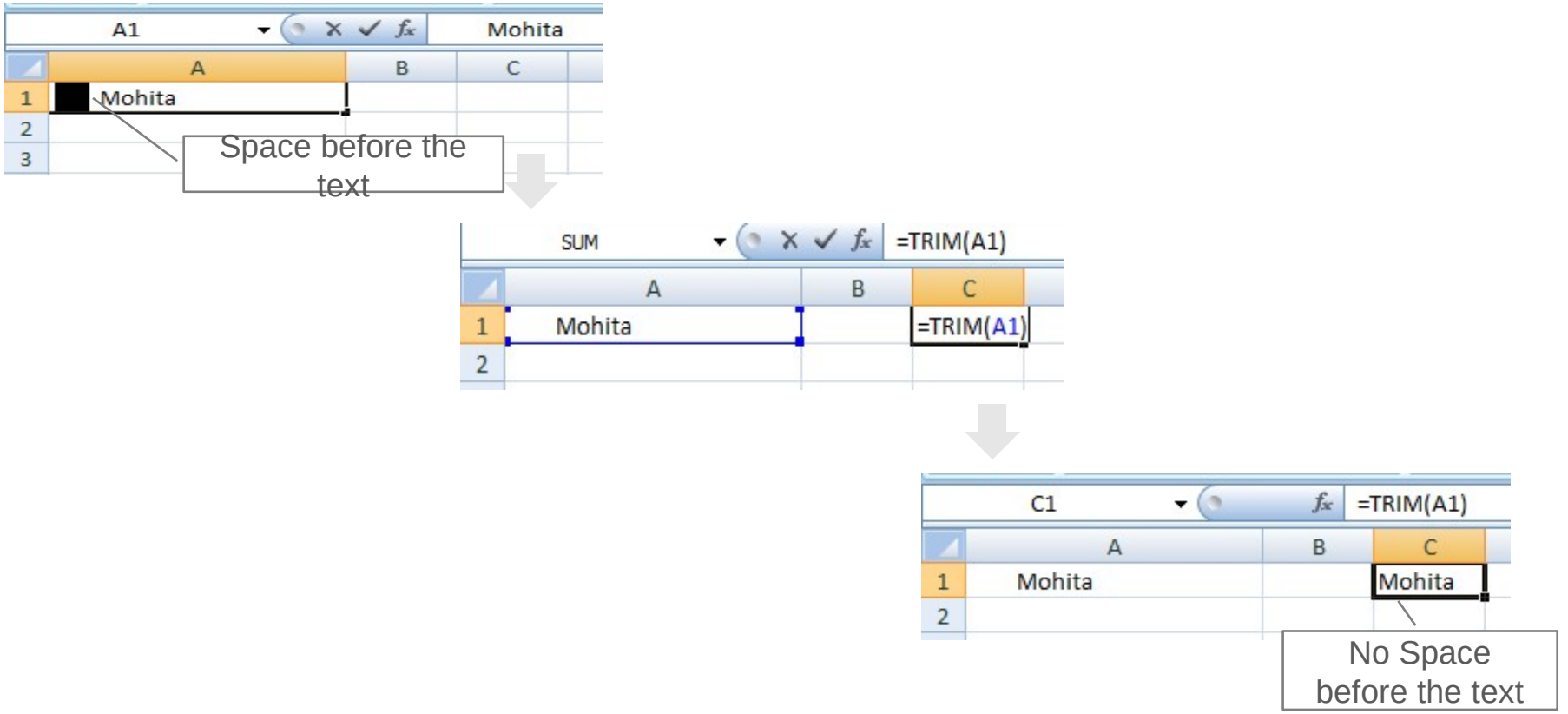


Tips:

If “num_chars” are omitted in the LEFT or RIGHT functions, it is assumed to be 1.

CHARACTER FUNCTIONS - TRIM

Function	Output	Syntax
TRIM()	Removes all spaces from text except for single spaces between words	=TRIM(text)



OTHER KEY CHARACTER FUNCTIONS

IMARTICUS LEARNING

Function	Usage
clean	Removes Non printable Characters from the text
exact	Checks if two strings are same or not
len	Returns the length of the text
lower	Helps in converting the Case of text into Lower Case
proper	Helps in converting the Case of text into Upper Case
upper	Helps in converting the Case of text into Proper Case
mid	Helps in extracting characters from the middle of a text
replace	Replaces part of a string with a different text

IMARTICUS LEARNING

Keyboard Shortcuts



KEYBOARD SHORTCUTS - FORMATTING

Shortcut	Output
Alt+’	Display the style dialog box
Ctrl+Shift+~	General Num. Format
Ctrl+Shift+$	Currency Format
Ctrl+Shift+ %	Percentage Format
Ctrl+Shift+ !	Comma Format
Ctrl+Shift+ &	Outline border
Ctrl+Shift+ _	Remove border
Ctrl+b	Bold
Ctrl+i	Italic
Ctrl+u	Underline
Ctrl+9	Hide rows

KEYBOARD SHORTCUTS - SELECTION

Shortcut	Output
Ctrl+Shift+9	Unhide rows
Ctrl+0	Hide columns
Ctrl+Shift+0	Unhide columns
Ctrl+1	Format Dialog Box
Ctrl+5	Strike Through
Ctrl+Space	Select entire row
Shift+ Space	Select entire column
Ctrl+a	Select entire worksheet
Ctrl+x/c/v	Cut/Copy/Paste

KEYBOARD SHORTCUTS - SELECTION

IMARTICUS LEARNING

Shortcut	Output
Ctrl+d/r	File cells down/right
Ctrl+Shift+*	Select current region around the active cell (active region comprise of blank rows and columns around the active cell)
Shift+ arrowkey	Extend the selection by one cell
Ctrl+Shift+ arrowkey	Extend the selection to last non blank cell in the same row or column as the active cell
Shift+home	Extend the selection to beginning of row
Ctrl+Shift+home	Extend the selection to beginning of worksheet
Ctrl+Shift+end	Extend the selection to the last cell used on the worksheet (lower-right corner)

KEYBOARD SHORTCUTS - WINDOWS OR WORKBOOK KEYS

IMARTICUS LEARNING

Shortcut	Output
Ctrl+F4	Closes workbook window
Alt+F4	Closes Excel
Ctrl+F10	Maximizes the workbook
Ctrl+F9	Minimizes the workbook
Ctrl+F5	Restore window size
F6	Next Pane
Shift+F6	Previous Pane
Ctrl+F6	Next window
Ctrl+Tab	Next window

KEYBOARD SHORTCUTS - WINDOWS OR WORKBOOK KEYS

Shortcut	Output
Shift+F11	Inserts a new sheet
Ctrl+s	Saves the workbook
F12	Saves as
Ctrl+o	Opens a workbook
Ctrl+n	Opens a new workbook
CTRL + Page Up/Page Down	Go to previous/next worksheet
CTRL + Home/End	Go to the first/last cell of the worksheet
CTRL + arrow key	Go to the next non empty cell



IMARTICUS LEARNING

Pivot Tables

ADVANCED EXCEL TOOLS - PIVOT TABLES

- Pivot table is an interactive worksheet table
  - Provides a powerful tool for summarizing large amounts of tabular data
- Similar to a cross-tabulation table
  - A pivot table classifies numeric data in a list based on other fields in the list
- General purpose:
  - Quickly summarize data from a worksheet or from an external source
  - Calculate totals, averages, counts, etc. based on any numeric fields in your table
  - Generate charts from your pivot tables

ADVANCED EXCEL TOOLS - PIVOT TABLES



Advantages

- Interactive: easily rearrange them by moving, adding, or deleting fields
- Dynamic: results are automatically recalculated whenever fields are added or dropped, or whenever categories are hidden or displayed
- Easy to update: “refreshable” if the original worksheet data changes

Pivot Tables – Setting It Up

- Data arranged in a list:
  - Columns represent fields
  - Rows represent a record of related data
- First row = column label
- Columns contain one sort of data
  - For example, text in one column and numeric values in a separate column
- Remove subtotals



PIVOT TABLES – SETTING IT UP

First	Last	Gender
Sandra	Archer	F
Hamilton	Paws	M

First	Last	Gender
Sandra	Archer	F
Sandra	Archer	F
Hamilton	Paws	M
Hamilton	Paws	M

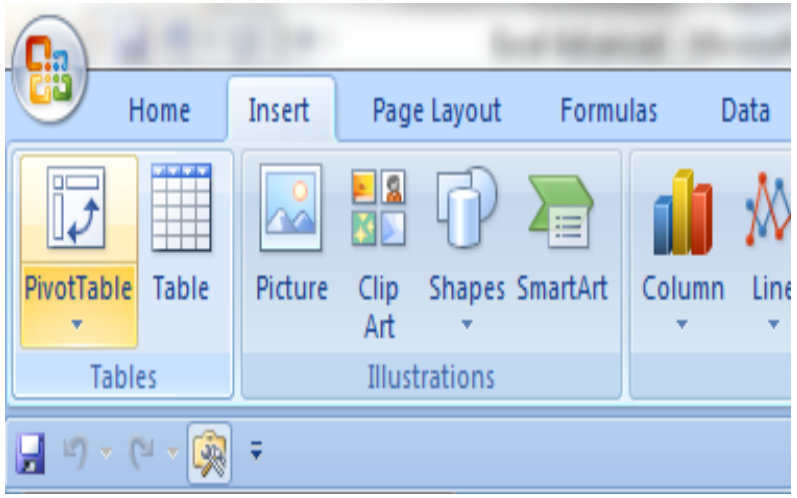
Incomplete records should be completed

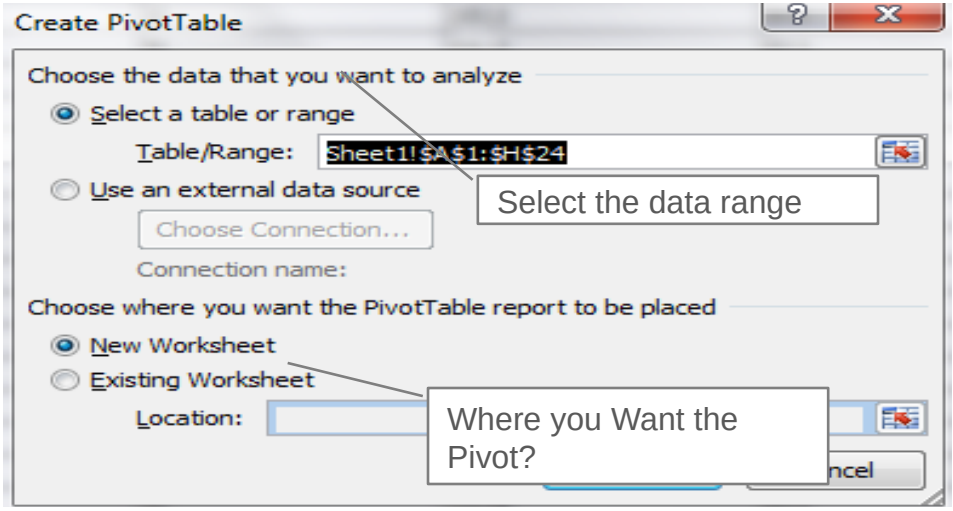
YTD Expenses		
2004	2005	2006
$4,000	$5,000	$6,000
$3,500	$4,500	$5,500
$2,000	$3,000	$0

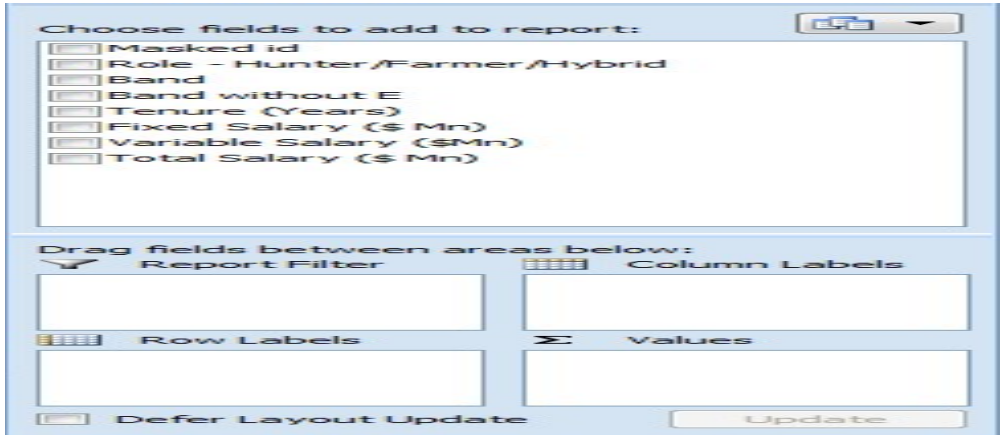
YTD Expense 2004	YTD Expense 2005	YTD Expense 2006
$4,000	$5,000	$6,000
$3,500	$4,500	$5,500
$2,000	$3,000	

Column Labels to be unique & Unmerged

PIVOT TABLES – SETTING IT UP







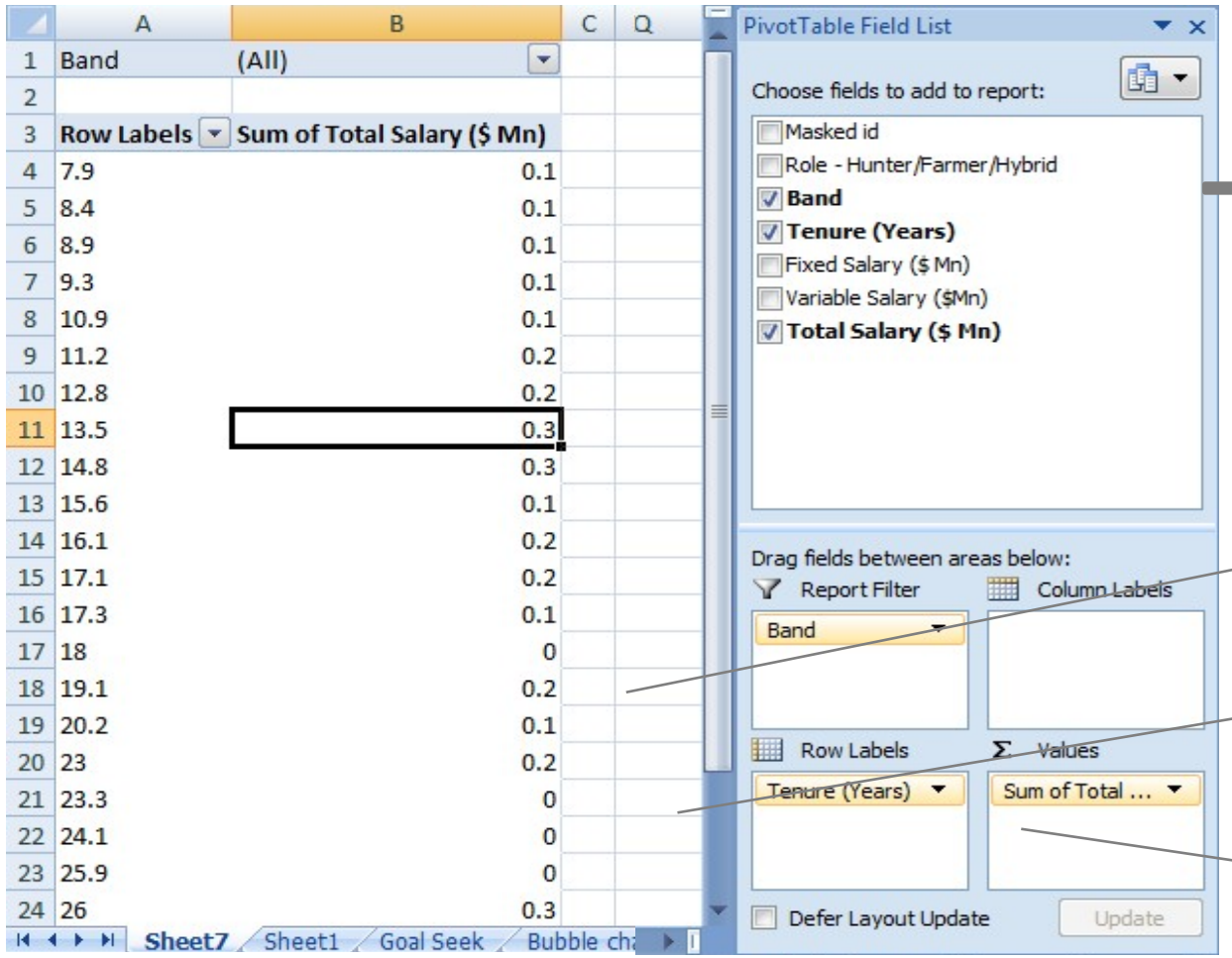
PIVOT TABLES – SETTING IT UP

- Remember the rules of where to place data fields:
  - **Row Fields:** display data vertically, in rows
  - **Column Fields:** display data horizontally, across columns
  - **Values Items:** numerical data to be summarized
  - **Report Filters:** Filter for data
- Changing the layout takes only seconds, so don’t worry about making it perfect the first time
- Note: If the field list is hidden, click Show Field List on the PivotTable toolbar



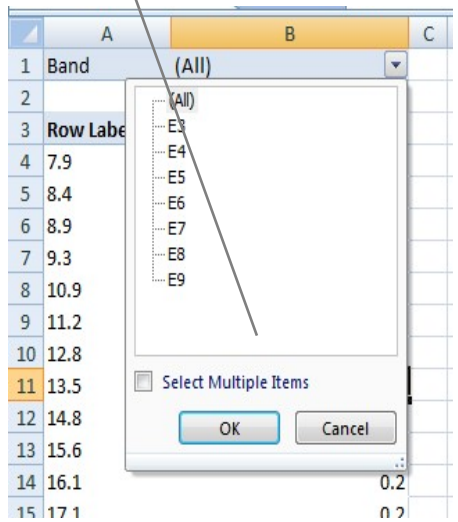
PIVOT TABLES – SETTING IT UP

- Create Pivot which gives Total Salary for each tenure year
- The report should have filtering capacity by Band



	A	B
1	Band	(All)
2		
3	Row Labels	Sum of Total Salary ($ Mn)
4	7.9	0.1
5	8.4	0.1
6	8.9	0.1
7	9.3	0.1
8	10.9	0.1
9	11.2	0.2
10	12.8	0.2
11	13.5	0.3
12	14.8	0.3
13	15.6	0.1
14	16.1	0.2
15	17.1	0.2
16	17.3	0.1
17	18	0
18	19.1	0.2
19	20.2	0.1
20	23	0.2
21	23.3	0
22	24.1	0
23	25.9	0
24	26	0.3

Sheet7 | Sheet1 | Goal Seek | Bubble ch



Filters allows to select bands

Filters required on band

Tenure is required as Row labels

Data required is salary

IMARTICUS LEARNING

Working with Formulae in Excel

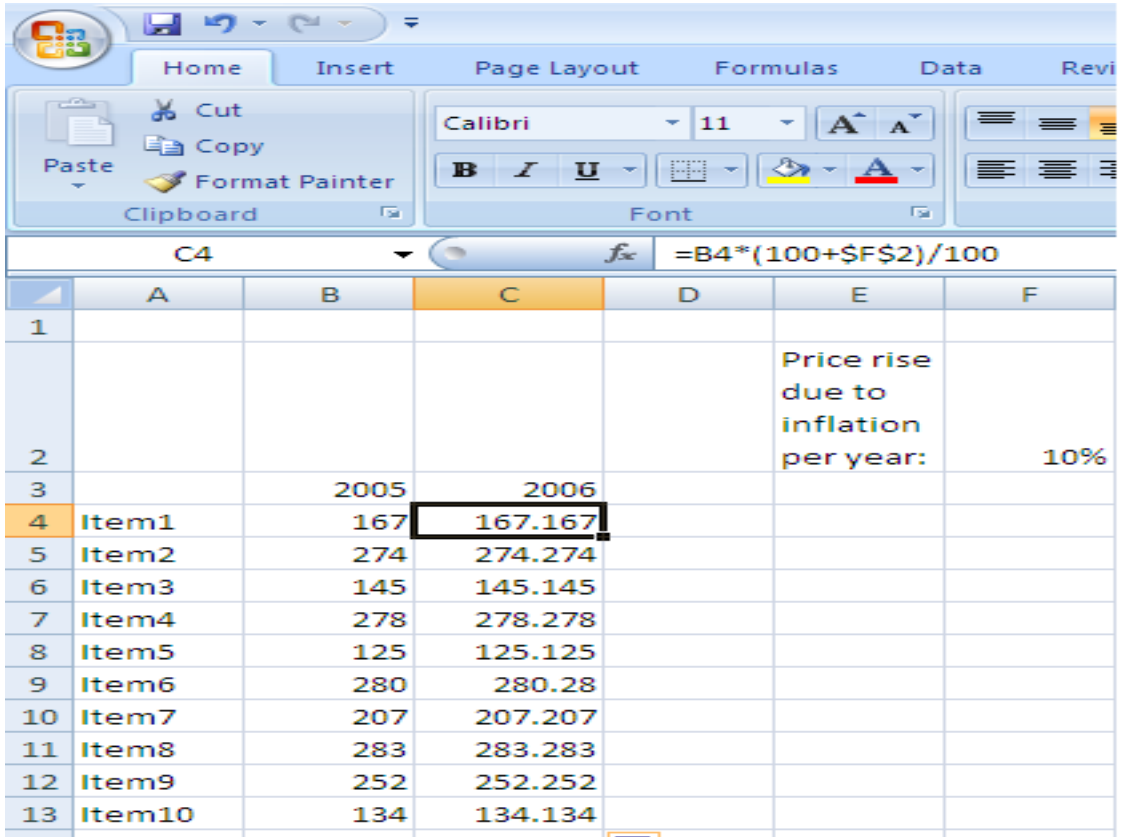


CELL REFERENCING – ABSOLUTE AND DYNAMIC REFERENCE

- To fix a cell value we put a “$” sign before both its column index and row index For ex. To fix cell value of cell “F2” throughout operations via dragging simply use “F2” as the cell reference instead of “F2”
- To fix the column just put the “$” sign before the column index. For “F2” it would be “$F2”
- To fix the row just put the “$” sign before the row index. For “F2” it would be “F$2”



CELL REFERENCING – ABSOLUTE AND DYNAMIC REFERENCE

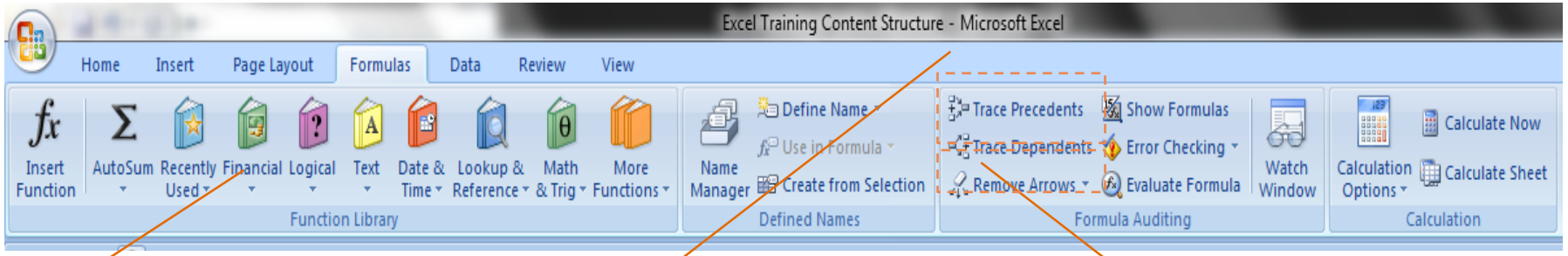


	A	B	C	D	E	F
1						
2					Price rise due to inflation per year:	10%
3		2005	2006			
4	Item1	167	167.167			
5	Item2	274	274.274			
6	Item3	145	145.145			
7	Item4	278	278.278			
8	Item5	125	125.125			
9	Item6	280	280.28			
10	Item7	207	207.207			
11	Item8	283	283.283			
12	Item9	252	252.252			
13	Item10	134	134.134			

In the adjacent example we wish to find out the expected price of each item in 2006, when the prices of previous year and an average raise of prices due to inflation.

Thus we fix the cell value of “F2” and calculate the expected prices as shown in the formula bar adjacently.

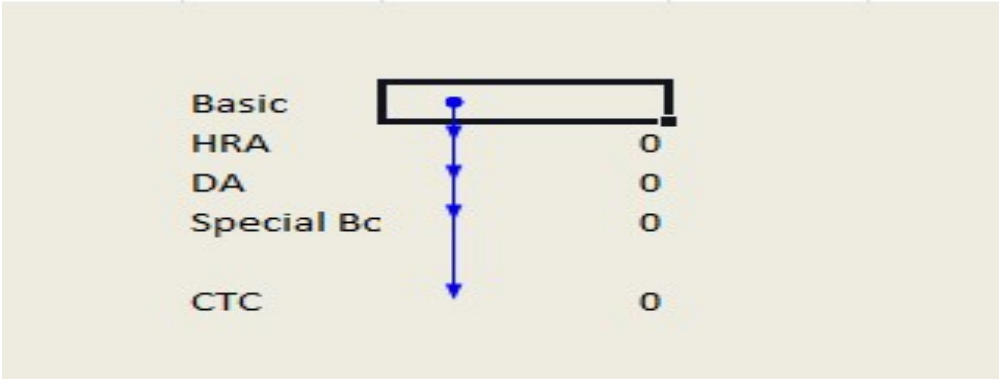
FORMULA AUDIT –(1/2)



List of Formulas by Categorization

For every cell, to see what all cells are dependent/ precedents

Remove all tracing arrows

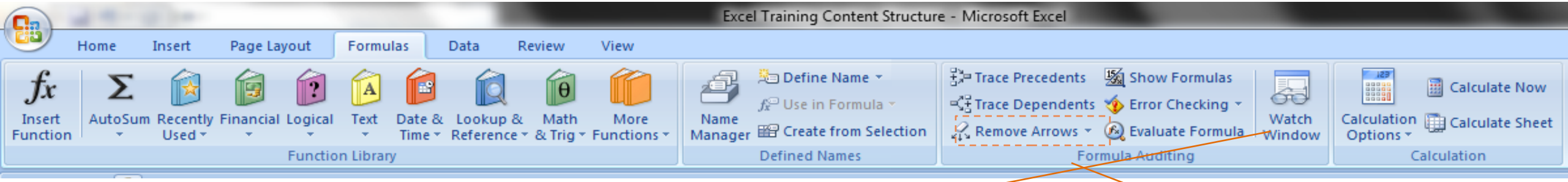


Before deleting any cell with formulas, check the dependents and assess the impact.

Caution: the dependent may be on other sheets hence always use this formula and not manual scrolling

Formula Audit – (2/2)

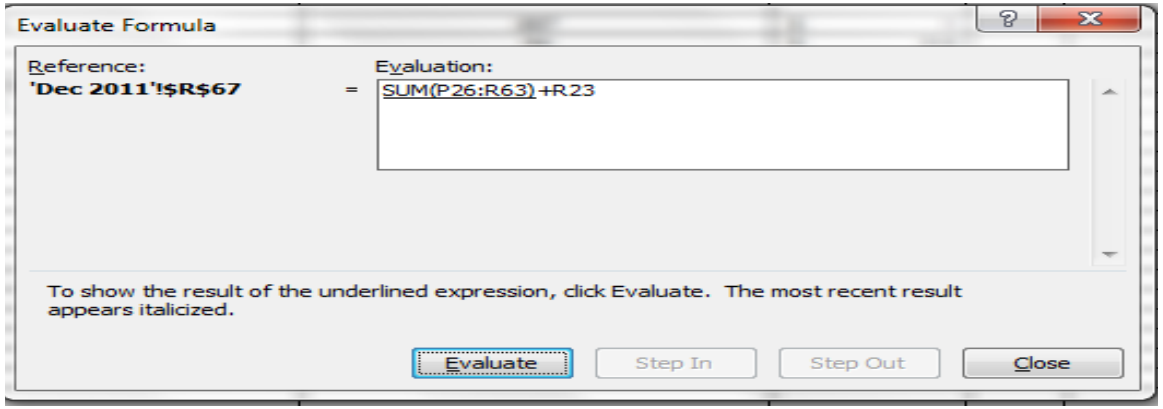
In case your formula is not giving expected results, use this to see step by step calculation



Displays the formula in each cell

Step by Step evaluation of formula

Click on Evaluate to see step by step results





IMARTICUS LEARNING

Working with Multiple Data Elements

VLOOKUP AND HLOOKUP

VLOOKUP and HLOOKUP allows you to find a specific cell of data in a larger data range.

- Use VLOOKUP when each row contains a separate record and the associated columns contain data for that one record.
- Use HLOOKUP when each column contains a separate record.



VLOOKUP AND HLOOKUP

VLOOKUP

VLOOKUP(SearchValue,Range,ColumnNumber,Error) ==> look for a value in the row specified by SearchValue and the column specified by Column Number

- SearchValue indicates the “match key” (i.e., find the row that contains the SearchValue in the first column)
- Range specifies the cells containing the data.
- Column Number specifies the column that contains the data element you want.
- Error determines what happens when Excel does not find the exact SearchValue you want. FALSE leads Excel to display a #N/A when an exact match cannot be found. TRUE leads Excel to display the next smaller value than

NOTE: *The 1st column of data must be sorted in ascending order when using VLOOKUP*

VLOOKUP AND HLOOKUP

Syntax: Hlookup



HLOOKUP(SearchValue,Range,RowNumber,Error) ==> look for a value in the column specified by SearchValue and the row specified by Row Number.

NOTE: *1st row of data must be sorted if using HLOOKUP*

EXAMPLE OF VLOOKUP AND HLOOKUP

	A	B	C	D	E	F
1	**Item Number**	**Shoe Type**	**Price**	**Discount**	**VLOOKUP ITEM NUMBER**	583
2	156	Slingbacks	£48.00	5%	**VLOOKUP SHOE TYPE**	
3	229	Stacked heel	£70.00	15%	**VLOOKUP PRICE**	
4	312	Stiletto	£85.00	20%	**VLOOKUP DISCOUNT PRICE**	
5	471	Platforms	£30.00	20%		
6	583	Kitten heel	£57.00	10%		
7	612	Mules	£40.00	15%		
8	773	Sandals	£37.00	5%		
9	823	Wedge	£35.00	5%		
10	982	Pumps	£45.00	5%		
11						

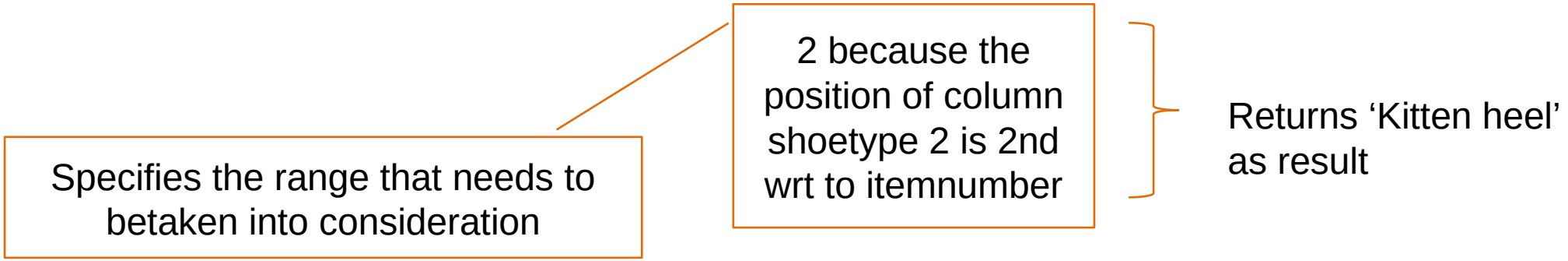
EXAMPLE OF VLOOKUP AND HLOOKUP

The item number we want to look up is in cell F1, and is a value of 583. We can use VLOOKUP to return the Shoe Type associated with item number 583.
Click inside cell F2 to select it, then click inside the formula bar at the top. Enter the following formula:

=VLOOKUP(F1, A2:C10, 2, FALSE)



EXAMPLE OF VLOOKUP AND HLOOKUP



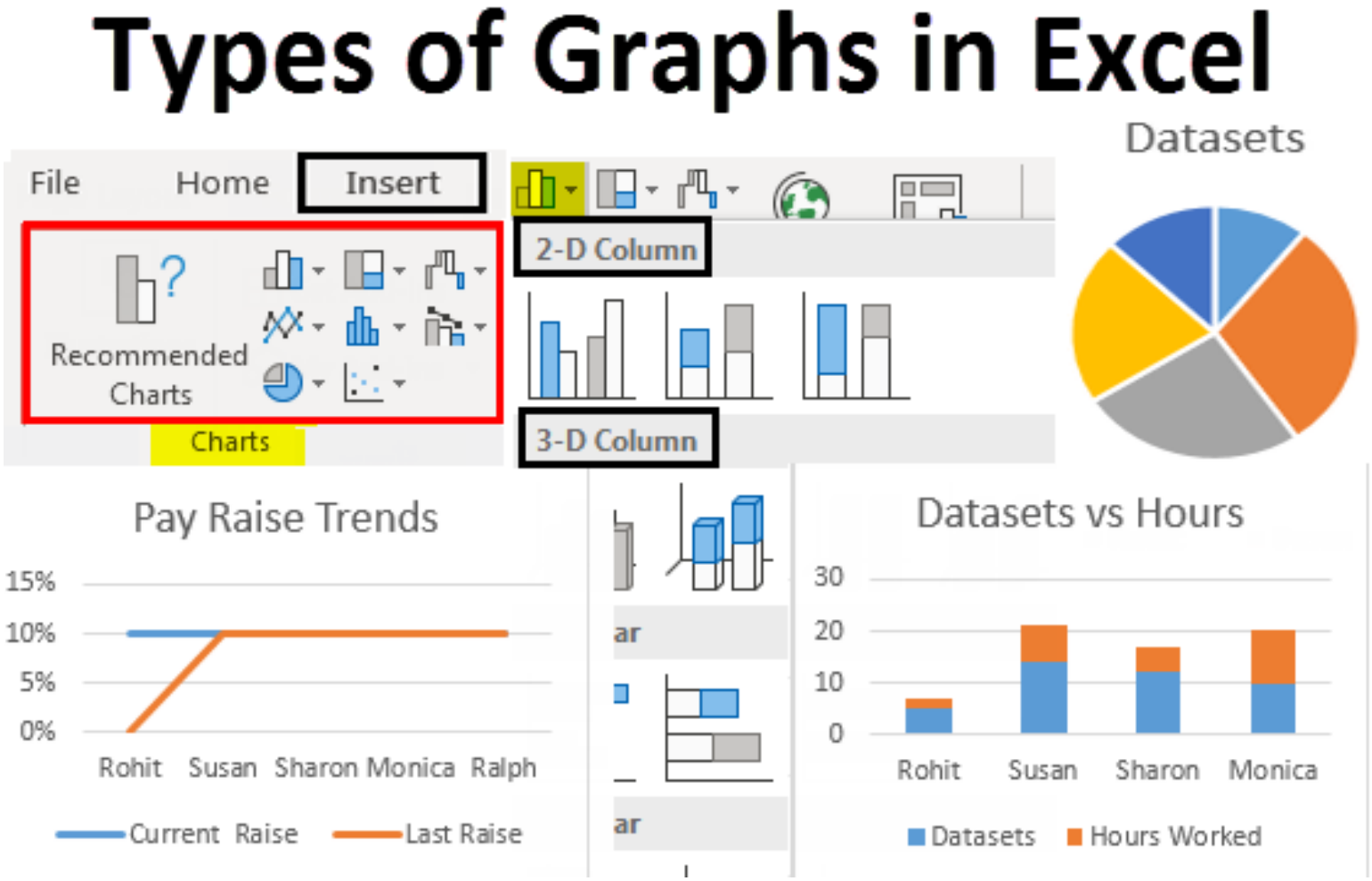
Tip:If “range_lookup” in VLOOKUP() or HLOOKUP() is TRUE, then the values in the first column of table_array must be placed in ascending order; otherwise VLOOKUP may not give the correct value. If range_lookup is FALSE, table_array does not need to be sorted.

IMARTICUS LEARNING

Charts and Graphs in Excel

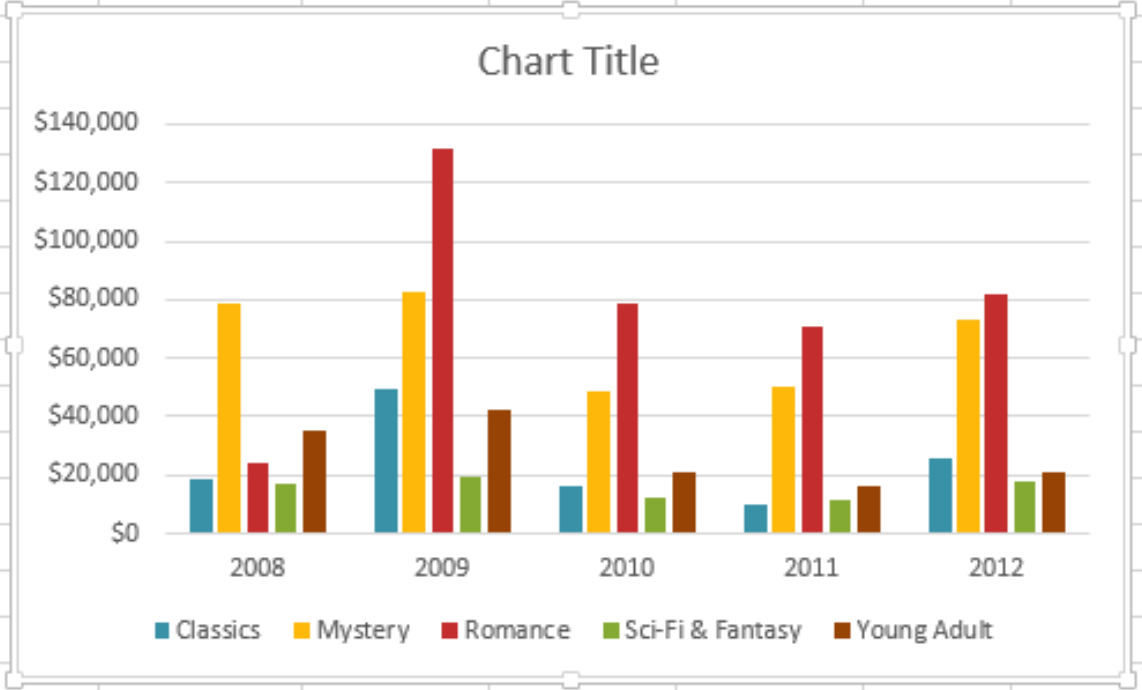


Charts and Graphs



Charts and Graphs

Genre	2008	2009	2010	2011	2012
Classics	$18,580	$49,225	$16,326	$10,017	$26,134
Mystery	$78,970	$82,262	$48,640	$49,985	$73,428
Romance	$24,236	$131,390	$79,022	$71,009	$81,474
Sci-Fi & Fantasy	$16,730	$19,730	$12,109	$11,355	$17,686
Young Adult	$35,358	$42,685	$20,893	$16,065	$21,388





What-If Analysis in Excel

What-If Analysis

What-if analysis in Excel is used to test more than one value for a different formula on the basis of multiple scenarios.

For this, we must have data of such kind where, for a single parameter, we would have 2 or more values for comparison.

There are three different kinds of tolls in the What-if analysis. Those are:

1. Scenario manager: The scenario manager helps to analyze and find the results for different scenarios.

2. Goal Seek: Goal seek helps to find the input for the known output or required output.

3. Data table: Data table helps to get results of different inputs in row-wise and column-wise.

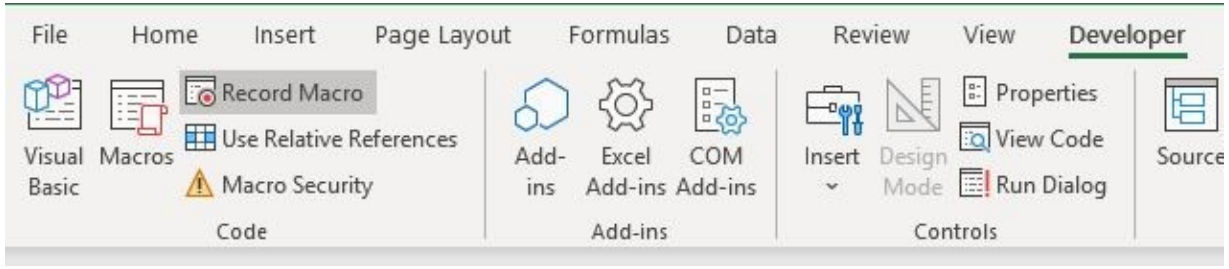
IMARTICUS LEARNING

Macros in Excel



Introduction to Macros

- Macros are a set of programming instructions written in VBA, which automate a repeated and standardized process in Excel and other Microsoft Office Applications.
- An Excel macro is an action or a set of actions that can be recorded, named, saved and executed as many times as required and whenever desired.
- By using macros, we are able to automate repetitive tasks associated with data manipulation and data reporting that must be accomplished repeatedly.
- An Excel Macro is a program that copies keystrokes or mouse actions repeated and common in the sheet.
- We can record/write macros in VBA and can run a macro by clicking the Macros command on the Developer tab on the ribbon.



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